

# Building and evaluation of a PBPK model for Dextromethorphan in adults with different CYP2D6 phenotypes

|                                                           |                                                                                                                                                                                             |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version                                                   | main-OSP12.2                                                                                                                                                                                |
| based on <i>Model Snapshot</i> and <i>Evaluation Plan</i> | <a href="https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/releases/tag/vmain">https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/releases/tag/vmain</a> |
| OSP Version                                               | 12.2                                                                                                                                                                                        |
| Qualification Framework Version                           | 3.5                                                                                                                                                                                         |

This evaluation report and the corresponding PK-Sim project file are filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/>

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# 1 Introduction

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Dextromethorphan is a widely used over-the-counter cough suppressant and a common ingredient of cold medicines marketed toward children and adults.

Dextromethorphan is typically administered as its hydrobromide salt, which is considered a Biopharmaceutics Drug Disposition Classification System (BDDCS) class I drug with high solubility and permeability. After oral administration, dextromethorphan is rapidly absorbed. Dextromethorphan undergoes an extensive first-pass metabolism, predominately mediated by CYP2D6, reducing the bioavailability to 1%–2% in CYP2D6 extensive metabolizers (EMs) and 80% in CYP2D6 poor metabolizers (PMs).

The herein presented model building and evaluation report evaluates the performance of the PBPK model for Dextromethorphan in (healthy) adults published by Rüdesheim et al. 2025 ([Rüdesheim 2025](#)). A whole-body PBPK model of dextromethorphan and its metabolites dextrorphan and dextrorphan O-glucuronide was developed and evaluated to predict drug plasma concentrations over a wide dosing range (5–80mg). Dextromethorphan is mainly metabolized via CYP2D6 and to a lesser extent by CYP3A4. A CYP2D6 activity score-dependent metabolism of dextromethorphan was implemented to describe the effect of CYP2D6 drug-gene interaction (DGI) on the PK of the modeled compounds. Although dextromethorphan is rapidly absorbed from the intestine, time to reach peak plasma concentration  $C_{max}$  ( $t_{max}$ ) often occurs as late as 4h after oral administration. This phenomenon likely occurs due to lysosomal trapping of dextromethorphan in the intestinal mucosa. To emulate the effect of lysosomal trapping in the gastrointestinal mucosa, a surrogate protein binding partner was included in the model.

Model features include:

Dextromethorphan:

- metabolism by CYP2D6 and formation of the primary metabolite dextrorphan
- metabolism by CYP3A4
- passive glomerular filtration
- lysosomal trapping mimicked by surrogate protein binding

Primary metabolite Dextrorphan:

- metabolism by UGT2B15 into Dextrorphan-O-glucuronide
- metabolism by CYP3A4

Secondary metabolite Dextrorphan-O-glucuronide:

- passive glomerular filtration
- active secretion to urine

The presented Dextromethorphan PBPK model as well as the respective evaluation plan and evaluation report are provided open-source (<https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model>).

## 2 Methods

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### 2.1 Modeling strategy

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The general concept of building a PBPK model has previously been described by Kuepfer et al. ([Kuepfer 2016](#)) Regarding the relevant anthropometric (height, weight) and physiological parameters (e.g. blood flows, organ volumes, binding protein concentrations, hematocrit, cardiac output) in adults was gathered from the literature and has been previously published ([PK-Sim Ontogeny Database Version 7.3](#)). The information was incorporated into PK-Sim® and was used as default values for the simulations in adults.

The applied activity and variability of plasma proteins and active processes that are integrated into PK-Sim® are described in the publicly available PK-Sim® Ontogeny Database Version 7.3 ([Schlender 2016](#)) or otherwise referenced for the specific process.

The dextromethorphan PBPK model was developed using a total of 28 clinical studies where dextromethorphan was administered as an intravenous infusion (one study), orally in single (26 studies), or multiple doses (one study), alone (17 studies) or as part of a phenotyping cocktail (11 studies). Doses ranged between 5 and 80 mg of administered dextromethorphan. The PBPK model was built based on data from healthy individuals, using the reported sex, ethnicity and mean values for age, weight and height from each study protocol. If no demographic information was provided, the following default values were substituted: male, European, 30 years of age, 73 kg body weight and 176 cm body height (characteristics from the PK-Sim® population database). CYP2D6 was implemented in accordance with literature, using the PK-Sim® expression database to define their relative expression in the different organs of the body. First a base model was built using clinical data from a study where dextromethorphan was administered as an intravenous infusion ([Duedahl 2004](#)) and oral administration from 3 clinical studies. The base model was further verified using 12 additional clinical studies. Next, the model was further developed using a DGI training dataset consisting of four studies that reported CYP2D6 activity scores or genotypes of their respective study populations. To complement these studies, 24 individual plasma concentration-time profiles were included. Overall, activity scores in the DGI model training dataset ranged from 0 (poor metabolizer PM) to 3 (ultrarapid metabolizer UR) and covered a total of eight activity scores. This dataset was used to optimize population *k<sub>cat</sub>* values for the activity scores of the respective studies or individual profiles

Unknown parameters (see below) were identified using the Parameter Identification module provided in PK-Sim®. Structural model selection was mainly guided by visual inspection of the resulting description of data and biological plausibility.

Once the appropriate structural model was identified, additional parameters for tablet formulations were identified.

The model was verified by simulating further clinical studies reporting pharmacokinetic concentration-time profiles after oral administration of dextromethorphan (verification datasets).

Details about input data (physicochemical, *in vitro* and clinical) can be found in [Section 2.2](#).

Details about the structural model and its parameters can be found in [Section 2.3](#).

### 2.2 Data and parameters used

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#### 2.2.1 In vitro / physico-chemical Data Dextromethorphan

A literature search was performed to collect available information on physicochemical properties of dextromethorphan. The obtained information from literature is summarized in the table below and is used for model building.

| Parameter                           | Unit   | Value  | Source                         | Description                                     |
|-------------------------------------|--------|--------|--------------------------------|-------------------------------------------------|
| MW                                  | g/mol  | 271.41 | <a href="#">Benet 2011</a>     | Molecular weight                                |
| pK <sub>a</sub>                     |        | 9.10   | <a href="#">Spaggiari 2014</a> | Acid dissociation constant                      |
| Solubility (pH)                     | g/L    | 15.00  | <a href="#">Benet 2011</a>     | Aqueous Solubility Hydrobromide                 |
| logP                                |        | 4.10   | <a href="#">Spaggiari 2014</a> | Partition coefficient between octanol and water |
| fu                                  | %      | 35.00  | <a href="#">Lutz 2012</a>      | Fraction unbound in plasma                      |
| CYP2D6 K <sub>m</sub> -> dxt        | μmol/L | 4.65   | <a href="#">Brown 2007</a>     | Michaelis Menten constant                       |
| CYP2D6 K <sub>cat</sub> -> dxt (PM) | 1/min  | 0      | <a href="#">Brown 2007</a>     | Catalytic rate constant                         |
| CYP3A4 K <sub>m</sub>               | μmol/L | 176.80 | <a href="#">Lutz 2012</a>      | Michaelis Menten constant                       |

## 2.2.2 In vitro / physico-chemical Data Dextrorphan

A literature search was performed to collect available information on physicochemical properties of Dextrorphan. The obtained information from literature is summarized in the table below and is used for model building.

| Parameter                           | Unit   | Value  | Source                              | Description                                     |
|-------------------------------------|--------|--------|-------------------------------------|-------------------------------------------------|
| MW                                  | g/mol  | 257.37 | <a href="#">HMDB-a,Wishart 2018</a> | Molecular weight                                |
| logP                                |        | 1.38   | <a href="#">Spaggiari 2014</a>      | Partition coefficient between octanol and water |
| pK <sub>a</sub> strongest acidic    |        | 10.10  | <a href="#">Spaggiari 2014</a>      | Acid dissociation constant                      |
| logP                                |        | 2.90   | <a href="#">Kim 2019</a>            | Partition coefficient between octanol and water |
| fu                                  | %      | 42.00  | <a href="#">Watanabe 2018</a>       | Fraction unbound in plasma                      |
| UGT2B15 K <sub>m</sub> -> dxt-glu   | μmol/L | 184.8  | <a href="#">Lutz 2012</a>           | Michaelis Menten constant                       |
| CYP2D6 K <sub>cat</sub> -> dxt (PM) | 1/min  | 0      | <a href="#">Brown 2007</a>          | Catalytic rate constant                         |
| CYP3A4 K <sub>m</sub>               | μmol/L | 910.00 | <a href="#">Lutz 2012</a>           | Michaelis Menten constant                       |
| CYP3A4 K <sub>cat</sub>             | 1/min  | 7.41   | <a href="#">Lutz 2012</a>           | Catalytic rate constant                         |

## 2.2.3 In vitro / physico-chemical Data Dextrorphan O-glucuronide

A literature search was performed to collect available information on physicochemical properties of Dextrorphan O-glucuronide. The obtained information from literature is summarized in the table below and is used for model building.

| Parameter                        | Unit  | Value | Source                                                | Description                                     |
|----------------------------------|-------|-------|-------------------------------------------------------|-------------------------------------------------|
| MW                               | g/mol | 433.5 | <a href="#">Wishart 2018</a>                          | Molecular weight                                |
| logP                             |       | 1.38  | <a href="#">HMDB-b</a> , <a href="#">Wishart 2018</a> | Partition coefficient between octanol and water |
| pK <sub>a</sub> strongest basic  |       | 9.82  | <a href="#">Wishart 2018</a>                          | Acid dissociation constant                      |
| pK <sub>a</sub> strongest acidic |       | 2.85  | <a href="#">Wishart 2018</a>                          | Acid dissociation constant                      |
| Solubility                       | g/L   | 1.20  | <a href="#">Wishart 2018</a>                          | Solubility                                      |
| fu                               | %     | 37.00 | Calculated <a href="#">Watanabe 2018</a>              | Fraction unbound in plasma                      |

## 2.2.4 Clinical Data

A literature search was performed to collect available clinical data on Dextromethorphan in healthy adults.

### 2.2.4.1 Base Model Building

The following studies were used for model building (training data):

| Publication                  | Study description                                                     |
|------------------------------|-----------------------------------------------------------------------|
| <a href="#">Duedahl 2005</a> | CYP2D6 EM Subjects with an single IV infusion of 0.5 mg/kg for 30 min |
| <a href="#">Schadel 1995</a> | CYP2D6 EM and PM Subjects with a single PO dose of 30 mg              |
| <a href="#">Tennezé 1990</a> | CYP2D6 EM subjects with a single PO dose of 80 mg                     |

### 2.2.4.2 Base Model Verification

The following studies were used for model verification:

| Publication                           | Study description                                         |
|---------------------------------------|-----------------------------------------------------------|
| <a href="#">Feld 2013</a>             | CYP2D6 EM subjects with a single PO dose of 60 mg         |
| <a href="#">AntecipBioventuresLLC</a> | CYP2D6 EM subjects with twice daily PO doses of 60 mg bid |
| <a href="#">Armani 2017</a>           | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Dumond 2010</a>           | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Edwards 2017</a>          | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Ermer 2015</a>            | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Kakuda 2014</a>           | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Khalilieh 2018</a>        | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Nakashima 2007</a>        | CYP2D6 EM subjects with a single PO dose of 50 mg sd      |
| <a href="#">Nyunt 2008</a>            | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Sager 2014</a>            | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |
| <a href="#">Stage 2018</a>            | CYP2D6 EM subjects with a single PO dose of 30 mg sd      |

### 2.2.4.3 DGI Model Building

The following studies were used for model building (training data):

| Publication              | Study description                                        |
|--------------------------|----------------------------------------------------------|
| <a href="#">Qiu 2016</a> | CYP2D6 NM and IM Subjects with a single PO dose of 15 mg |

### 2.2.4.4 DGI Model Verification

The following studies were used for model verification:

| Publication                     | Study description                                           |
|---------------------------------|-------------------------------------------------------------|
| <a href="#">Capon 1996</a>      | CYP2D6 EM and PM subjects with a single PO dose of 30 mg    |
| <a href="#">Gazzaz 2018</a>     | CYP2D6 NM subjects with a single PO dose of 30 mg sd        |
| <a href="#">Gorski 2004</a>     | CYP2D6 EM and PM subjects with a single PO dose of 30 mg sd |
| <a href="#">Yamazaki 2017</a>   | CYP2D6 NM and IM subjects with a single PO dose of 30 mg sd |
| <a href="#">Storelli 2018</a>   | CYP2D6 EM subjects with a single PO dose of 5 mg sd         |
| <a href="#">Zawertailo 2010</a> | CYP2D6 NM subjects with a single PO dose of 3 mg/kg sd      |

## 2.3 Model Parameters and Assumptions

### 2.3.1 Absorption

The dissolution of tablets were implemented via an empirical Weibull dissolution capsule. The final Weibull shape parameters and Weibull time parameters (50% dissolved) for the cocktail capsule formulation used in the PBPK model are given in [Section 2.3.4](#).

Although dextromethorphan is rapidly absorbed from the intestine, time to reach peak plasma concentration  $C_{max}$  ( $t_{max}$ ) often occurs as late as 4h after oral administration. This phenomenon likely occurs due to lysosomal trapping of dextromethorphan in the intestinal mucosa. However, other processes, such as renal excretion may also be affected by lysosomal trapping in the respective tissue. In short, lipophilic amines ( $\log P > 1$ , acid dissociation constant ( $pK_a$ )  $> 6$ ) accumulate in lysosomes due to rapid diffusion across the lysosomal membrane in unionized form. Subsequently, due to the acidic environment in lysosomes (pH 4–5), the amine is then ionized and thus unable to permeate back into the cytosol. The information necessary to physiologically implement lysosomal trapping (i.e. relative abundances of lysosomes in relevant tissues and diffusion constants for permeation across lysosomal membranes) are not yet available in the literature. Hence, intestinal lysosomal trapping was implemented as follows: First, a surrogate protein binding partner was expressed in high abundances (500  $\mu\text{mol/L}$ ) in the relevant tissues (duodenum, upper jejunum, lower jejunum, upper ileum and lower ileum, each 100% of relative expression). Secondly, a corresponding protein binding process was implemented for dextromethorphan. Finally, the relevant parameters for the binding process - dissociation rate constant ( $k_{off}$ ) and dissociation constant ( $K_D$ ) - were informed by parameter optimization. For a comprehensive explanation on the process of lysosomal trapping under physiological circumstances, please refer to [Kazmi 2013](#). The final parameters for  $K_D$  and  $k_{off}$  are given in [Section 2.3.4](#).

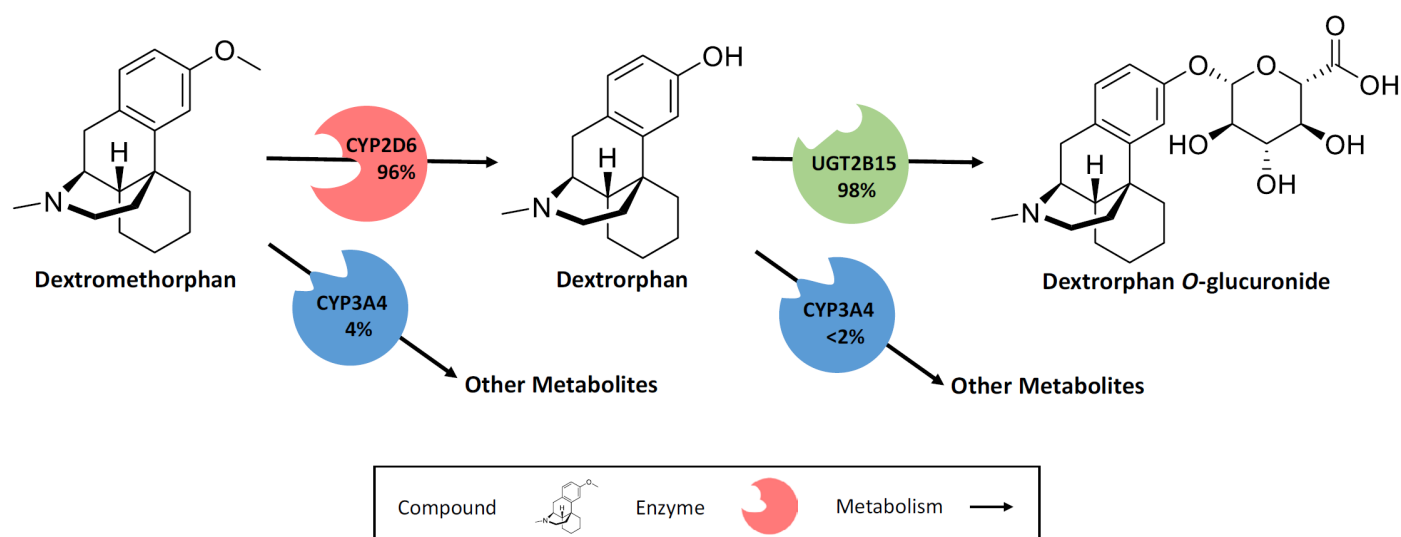
### 2.3.2 Distribution

After testing the available organ-plasma partition coefficient and cell permeability calculation methods built in PK-Sim, observed Dextromethorphan and dextrorphan clinical data was best described by choosing the partition coefficient calculation by [Berezhkovskiy](#) and cellular permeability calculation by [PK-Sim Standard](#). For the Dextrorphan-O-glucuronide clinical data was best described by choosing the partition coefficient calculation by [Berezhkovskiy](#) and cellular permeability calculation by [Charge dependent Schmitt](#).

### 2.3.3 Metabolism and Elimination

Dextromethorphan-O-demethylation via CYP2D6 leads to the formation of the major active metabolite dextrorphan. Dextrorphan subsequently undergoes rapid glucuronidation via uridine diphosphate-glucuronosyltransferases 2B (UGT2Bs), namely UGT2B15, or N-demethylation via CYP3A4. **Figure 1** shows the implemented metabolic pathways.

**Figure 1: Implemented dextromethorphan metabolic pathways.**



Alternatively, dextromethorphan is N-demethylated by CYP3A4, which was found to be the main pathway of dextromethorphan metabolism in CYP2D6 PMs. Depending on the CYP2D6 phenotype, up to 50% of orally administered dextromethorphan is excreted unchanged in urine. For dextromethorphan and Dextrorphan, a GFR fraction of 1 was assumed. Dextrorphan O-glucuronide is renally eliminated via passive glomerular filtration and active secretion to the urine, a GFR fraction > 1 was optimized and represents both processes [Section 2.3.4](#).

The principal pathway of dextromethorphan metabolism is the CYP2D6-mediated O-demethylation, leading to the formation of dextrorphan. This pathway was implemented using Michaelis-Menten kinetics. Cytochrome P450 2D6 (CYP2D6) was implemented in accordance with literature, using the PK-Sim® expression database to define their relative expression in the different organs of the body [PK-Sim Ontogeny Database Version 7.3](#). The CYP2D6, CYP3A4 and UGT2B15 expression profiles are based on high-sensitive real-time RT-PCR [Nishimura 2013](#). The  $t_{1/2}$  for CYP3A4 was decreased from the default 37h to 36h and the reference concentration for UGT2B15 was increased from 1  $\mu\text{mol/L}$  to 2.48  $\mu\text{mol/L}$  for a better fit of the clinical data.

Because the CYP2D6 gene is prone to genetic alterations, dextromethorphan pharmacokinetics is subject to considerable drug-gene interaction (DGI) effects. For DGI modeling, the CYP2D6 Michaelis-Menten constant ( $K_M$ ) values for the dextromethorphan O-demethylation were kept constant over the whole range of modeled activity scores [Rüdesheim 2020](#). CYP2D6  $k_{\text{cat}}$  values were optimized separately for each activity score. CYP2D6 poor metabolizers (PMs) (activity score = 0) were assumed to show no CYP2D6 activity (0%), whereas populations with two wildtype alleles (activity score = 2) were assumed to possess normal CYP2D6 activity (100%). Activity scores were assigned according to Caudle et al. 2017 [Caudle 2017](#), see also table below. IM = intermediate metabolizers, NM = normal metabolizers and UM = ultra-rapid metabolizer.

| Activity Score | Projected | $k_{\text{cat}} \rightarrow \text{dxt}$ (1/min) | $k_{\text{cat}}$ percentage of Reference (AS = 2) (%) |
|----------------|-----------|-------------------------------------------------|-------------------------------------------------------|
| 0.00           | PM        | 0.0                                             | 0                                                     |
| 0.25           |           | 5.3                                             | 2                                                     |
| 0.50           | IM        | 32.9                                            | 14                                                    |
| 1.00           |           | 96.6                                            | 40                                                    |
| 1.25           |           | 115.2                                           | 48                                                    |
| 1.50           | NM        | 151.8                                           | 63                                                    |
| 2.00           |           | 242.5                                           | 100                                                   |
| 3.00           | UM        | 413.2                                           | 170                                                   |

In addition, fraction of bile that was continuously released was assumed 1 ( EHC continuous fraction ) for Dextromethorphan and its metabolites.

## 2.3.4 Automated Parameter Identification

| Model Parameter                                      |
|------------------------------------------------------|
| <b>Dextromethorphan</b>                              |
| Weibull time parameter                               |
| Weibull shape parameter                              |
| Intestinal permeability                              |
| CYP2D6 $K_{\text{cat}} \rightarrow \text{dxt}$ (EM)  |
| CYP3A4 $K_{\text{cat}}$                              |
| Lysosomal trapping $K_D$                             |
| Lysosomal trapping $k_{\text{off}}$                  |
| <b>Dextrorphan</b>                                   |
| UGT2B15 $K_{\text{cat}} \rightarrow \text{dxt}$ (EM) |
| <b>Dextrorphan-O-glucuronide</b>                     |
| logP                                                 |
| GFR fraction                                         |

## 3 Results and Discussion

The PBPK model for Dextromethorphan, dextrorphan and dextrorphan O-glucuronide was developed and verified with clinical pharmacokinetic data.

The PBPK model for dextromethorphan was developed and verified with clinical pharmacokinetic data. The model was built and evaluated covering data from studies including in particular intravenous (0.5 mg/kg 30 min infusion ) oral administrations (dose range 5 to 80 mg).

The model quantifies metabolism of dextromethorphan via mainly CYP2D6, and includes the possibility to include  $K_{cat}$  activity scores for CYP2D6 to simulate different CYP2D6 phenotypes.

The next sections show:

1. the overall goodness of fit: [Section 3.2](#).
2. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: [Section 3.3](#).

### 3.1 Final input parameters

The compound parameter values of the final PBPK model are illustrated below.

#### Compound: Dextromethorphan

##### Parameters

| Name                                             | Value           | Value Origin                                                | Alternative | Default |
|--------------------------------------------------|-----------------|-------------------------------------------------------------|-------------|---------|
| Solubility at reference pH                       | 15 mg/ml        | Publication-Other-Benet 2011                                | Measurement | True    |
| Reference pH                                     | 7.4             | Publication-Other-Benet 2011                                | Measurement | True    |
| Lipophilicity                                    | 4.1 Log Units   | Publication-Other-Spaggiari 2014                            | LogP        | True    |
| Fraction unbound (plasma, reference value)       | 0.35            | Publication-Other-Taylor 2016                               | Measurement | True    |
| Specific intestinal permeability (transcellular) | 2.48E-06 cm/min | Parameter Identification-Parameter Identification-Optimized | Optimized   | True    |
| Is small molecule                                | Yes             |                                                             |             |         |
| Molecular weight                                 | 271.4 g/mol     | Unknown-Sager 2014                                          |             |         |
| Plasma protein binding partner                   | Unknown         |                                                             |             |         |

##### Calculation methods

| Name                    | Value           |
|-------------------------|-----------------|
| Partition coefficients  | Berezhkovskiy   |
| Cellular permeabilities | PK-Sim Standard |

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

| Name         | Value | Value Origin |
|--------------|-------|--------------|
| GFR fraction | 1     |              |

Metabolizing Enzyme: CYP2D6-Lutz 2012 (dextrophan)

Molecule: CYP2D6

Metabolite: Dextrophan

Parameters

| Name                                        | Value                          | Value Origin                                                |
|---------------------------------------------|--------------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0.134 nmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein        | Unknown                                                     |
| Km                                          | 4.65 µmol/l                    | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 90.89 1/min                    | Parameter Identification-Parameter Identification-Optimized |

Metabolizing Enzyme: CYP3A4-Brown 2007 (sink)

Molecule: CYP3A4

Parameters

| Name                               | Value                         | Value Origin                                                |
|------------------------------------|-------------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes | 0.61 nmol/min/mg mic. protein |                                                             |
| Km                                 | 176.8 µmol/l                  | Publication-In Vitro-Brown 2007                             |
| kcat                               | 7.98 1/min                    | Parameter Identification-Parameter Identification-Optimized |

Metabolizing Enzyme: CYP2D6-AS=0 (dextrophan)

Molecule: CYP2D6

Metabolite: Dextrorphan

Parameters

| Name                                        | Value                      | Value Origin |
|---------------------------------------------|----------------------------|--------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |              |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown      |
| Km                                          | 4.65 µmol/l                |              |

Metabolizing Enzyme: CYP2D6-AS=2 (dextrorphan)

Molecule: CYP2D6

Metabolite: Dextrorphan

Parameters

| Name                                        | Value                      | Value Origin                                      |
|---------------------------------------------|----------------------------|---------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                   |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                           |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                    |
| kcat                                        | 242.45 1/min               | Parameter Identification-Parameter Identification |

Metabolizing Enzyme: CYP2D6-AS=1.25 (dextrorphan)

Molecule: CYP2D6

Metabolite: Dextrorphan

Parameters

| Name                                        | Value                      | Value Origin                                                |
|---------------------------------------------|----------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                     |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 115.17 1/min               | Parameter Identification-Parameter Identification-Optimized |

Metabolizing Enzyme: CYP2D6-AS=0.5 (dextrorphan)

Molecule: CYP2D6

Metabolite: Dextrorphan

#### Parameters

| Name                                        | Value                      | Value Origin                                                |
|---------------------------------------------|----------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                     |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 32.91 1/min                | Parameter Identification-Parameter Identification-Optimized |

#### Metabolizing Enzyme: CYP2D6-AS=1 (dextrorphan)

Molecule: CYP2D6

Metabolite: Dextrorphan

#### Parameters

| Name                                        | Value                      | Value Origin                                                |
|---------------------------------------------|----------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                     |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 96.64 1/min                | Parameter Identification-Parameter Identification-Optimized |

#### Specific Binding: lysosomal trapping mucosa-lysosomal trapping (Kazmi 2013)

Molecule: lysosomal trapping mucosa

#### Parameters

| Name | Value           | Value Origin                                                |
|------|-----------------|-------------------------------------------------------------|
| koff | 709939.97 1/min | Parameter Identification-Parameter Identification-Optimized |
| Kd   | 74.21 µmol/l    | Parameter Identification-Parameter Identification-Optimized |

#### Metabolizing Enzyme: CYP2D6-AS=0.25 (dextrorphan)

Molecule: CYP2D6

Metabolite: Dextrorphan

#### Parameters

| Name                                        | Value                      | Value Origin                                                |
|---------------------------------------------|----------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                     |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 5.3 1/min                  | Parameter Identification-Parameter Identification-Optimized |

Metabolizing Enzyme: CYP2D6-AS=1.5 (dextrophan)

Molecule: CYP2D6

Parameters

| Name                                        | Value                      | Value Origin                                                                                                          |
|---------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                                                                                       |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                                                                               |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                                                                                        |
| kcat                                        | 151.81 1/min               | Parameter Identification-Parameter Identification-Value updated from 'Parameter Identification 6' on 2021-01-19 09:07 |

Metabolizing Enzyme: CYP2D6-AS=3 (dextrophan)

Molecule: CYP2D6

Parameters

| Name                                        | Value                      | Value Origin                                                |
|---------------------------------------------|----------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 0 pmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 10 pmol/mg mic. protein    | Unknown                                                     |
| Km                                          | 4.65 µmol/l                | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 413.19 1/min               | Parameter Identification-Parameter Identification-Optimized |

Formulation: capsule/solution

Type: Dissolved

Formulation: cocktail capsule (weibull)

Type: Weibull

Parameters

| Name                             | Value     | Value Origin                                                |
|----------------------------------|-----------|-------------------------------------------------------------|
| Dissolution time (50% dissolved) | 46.05 min | Parameter Identification-Parameter Identification-Optimized |
| Lag time                         | 0 min     |                                                             |
| Dissolution shape                | 1.05      | Parameter Identification-Parameter Identification-Optimized |
| Use as suspension                | Yes       |                                                             |

Compound: Dextrorphan

Parameters

| Name                                       | Value         | Value Origin                       | Alternative | Default |
|--------------------------------------------|---------------|------------------------------------|-------------|---------|
| Solubility at reference pH                 | 0.17 mg/ml    | Database-Other-metabocard/ChemAxon | Measurement | True    |
| Reference pH                               | 7.4           | Database-Other-metabocard/ChemAxon | Measurement | True    |
| Lipophilicity                              | 2.9 Log Units | Database-Other-metabocard/ChemAxon | LogP        | True    |
| Fraction unbound (plasma, reference value) | 0.42          | Publication-Other-Watanabe 2008    | Measurement | True    |
| Is small molecule                          | Yes           |                                    |             |         |
| Molecular weight                           | 257.37 g/mol  | Database-Other-metabocard/ChemAxon |             |         |
| Plasma protein binding partner             | Unknown       |                                    |             |         |

Calculation methods

| Name                    | Value           |
|-------------------------|-----------------|
| Partition coefficients  | Berezhkovskiy   |
| Cellular permeabilities | PK-Sim Standard |

Processes

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

| Name         | Value | Value Origin |
|--------------|-------|--------------|
| GFR fraction | 1     |              |

### Metabolizing Enzyme: CYP3A4-Lutz 2012 (sink)

Molecule: CYP3A4

#### Parameters

| Name                               | Value                        | Value Origin                                                |
|------------------------------------|------------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes | 0.8 nmol/min/mg mic. protein |                                                             |
| Km                                 | 910 µmol/l                   | Publication-In Vitro-Lutz 2012                              |
| kcat                               | 7.41 1/min                   | Parameter Identification-Parameter Identification-Optimized |

### Metabolizing Enzyme: UGT2B15-Lutz 2012 (Dextrorphan-O-glucuronide)

Molecule: UGT2B15

Metabolite: Dextrorphan-O-glucuronide

#### Parameters

| Name                                        | Value                        | Value Origin                                                |
|---------------------------------------------|------------------------------|-------------------------------------------------------------|
| In vitro Vmax for liver microsomes          | 2.3 nmol/min/mg mic. protein |                                                             |
| Content of CYP proteins in liver microsomes | 62.1 pmol/mg mic. protein    | Publication-In Vitro-Achour 2014                            |
| Km                                          | 184.8 µmol/l                 | Publication-In Vitro-Lutz 2012                              |
| kcat                                        | 1137.98 1/min                | Parameter Identification-Parameter Identification-Optimized |

## Compound: Dextrorphan-O-glucuronide

#### Parameters

| Name                                          | Value                     | Value Origin                                                                                                                | Alternative | Default |
|-----------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|---------|
| Solubility at reference pH                    | 1200 mg/l                 | Parameter Identification-Parameter Identification-Value updated from 'Parameter Identification 6 - sol' on 2021-02-26 11:12 | Measurement | True    |
| Reference pH                                  | 7                         |                                                                                                                             | Measurement | True    |
| Lipophilicity                                 | 0.2928540753<br>Log Units | Parameter Identification-Parameter Identification-Value updated from 'Parameter Identification 6 - sol' on 2021-02-26 11:12 | LogP        | True    |
| Fraction unbound<br>(plasma, reference value) | 0.366                     | Parameter Identification-Parameter Identification-Value updated from 'Parameter Identification 5' on 2020-10-21 11:17       | Measurement | True    |
| Is small molecule                             | Yes                       |                                                                                                                             |             |         |
| Molecular weight                              | 433.5 g/mol               |                                                                                                                             |             |         |
| Plasma protein binding<br>partner             | Albumin                   |                                                                                                                             |             |         |

### Calculation methods

| Name                    | Value                    |
|-------------------------|--------------------------|
| Partition coefficients  | Berezhkovskiy            |
| Cellular permeabilities | Charge dependent Schmitt |

### Processes

#### Systemic Process: Glomerular Filtration-Assumption

Species: Human

#### Parameters

| Name         | Value | Value Origin                                                |
|--------------|-------|-------------------------------------------------------------|
| GFR fraction | 4.92  | Parameter Identification-Parameter Identification-Optimized |

## 3.2 Dextromethorphan and its metabolites Diagnostics Plots

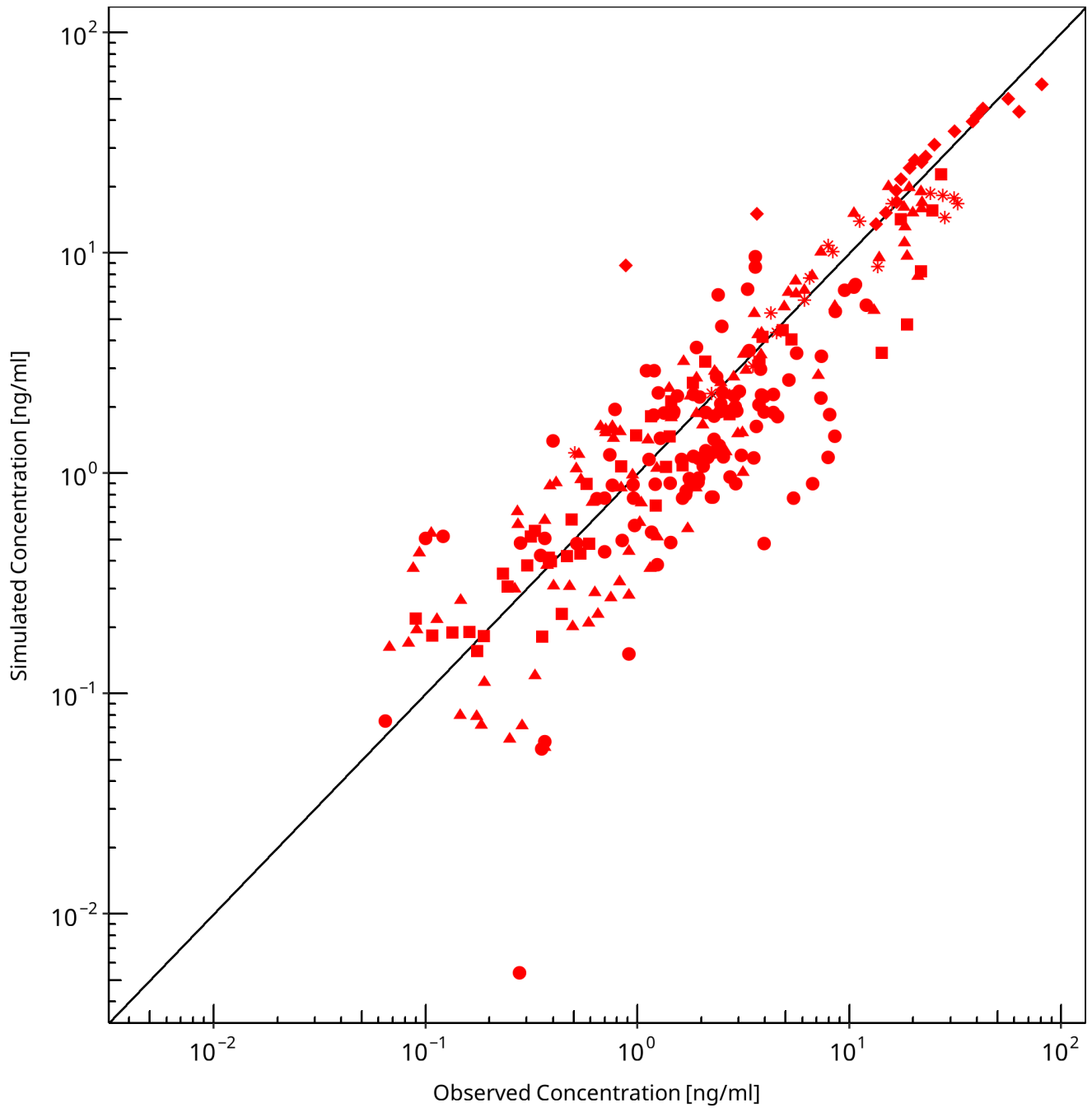
Below you find the goodness-of-fit visual diagnostic plots for the PBPK model performance of all data presented in [Section 2.2.2](#).

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

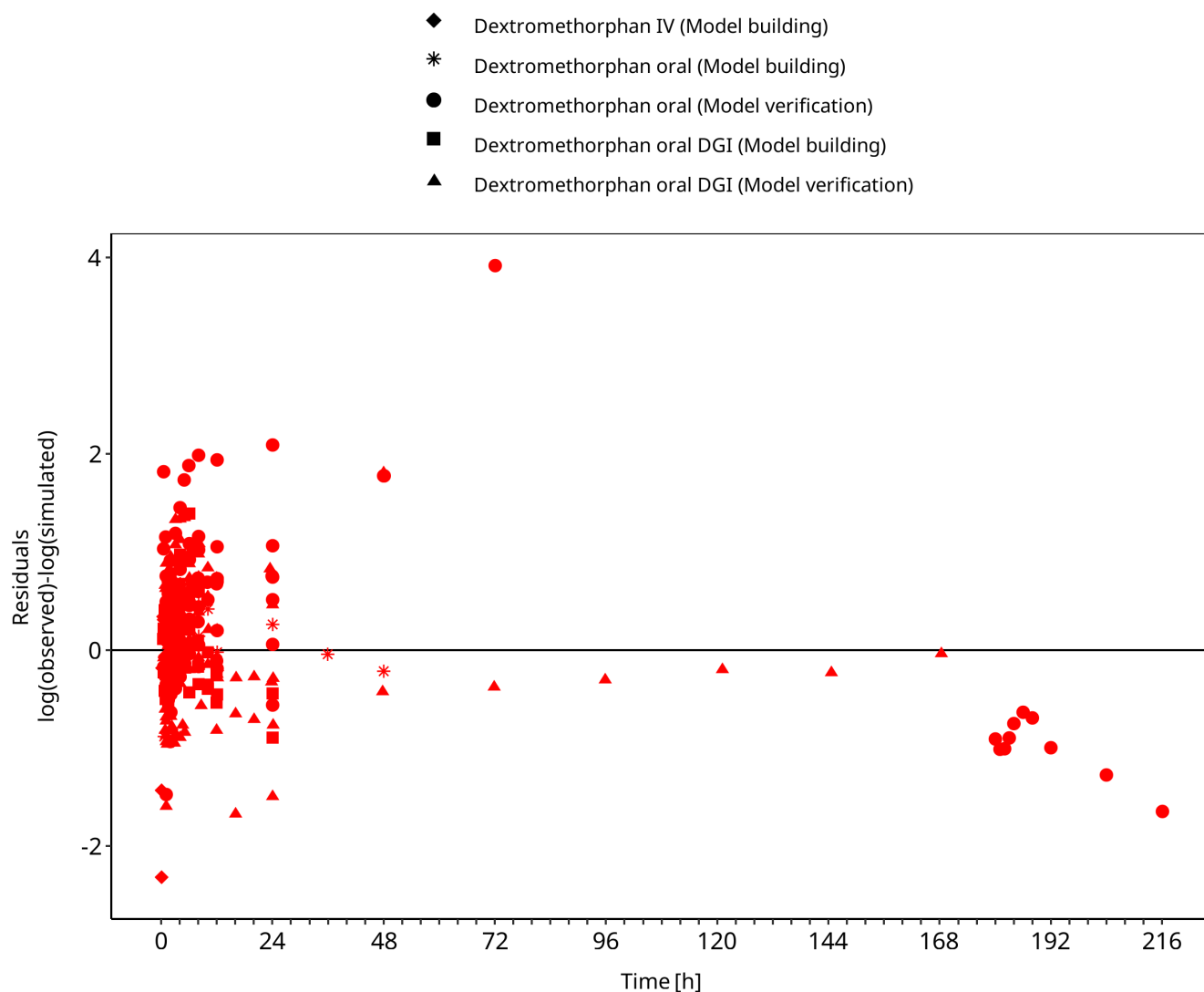
**Table 3-1: GMFE for Goodness of fit plot for Dextromethorphan concentration in plasma**

| Group                                          | GMFE |
|------------------------------------------------|------|
| Dextromethorphan IV (Model building)           | 1.40 |
| Dextromethorphan oral (Model building)         | 1.36 |
| Dextromethorphan oral (Model verification)     | 2.03 |
| Dextromethorphan oral DGI (Model building)     | 1.46 |
| Dextromethorphan oral DGI (Model verification) | 1.77 |
| All                                            | 1.75 |

- ◆ Dextromethorphan IV (Model building)
- \* Dextromethorphan oral (Model building)
- Dextromethorphan oral (Model verification)
- Dextromethorphan oral DGI (Model building)
- ▲ Dextromethorphan oral DGI (Model verification)



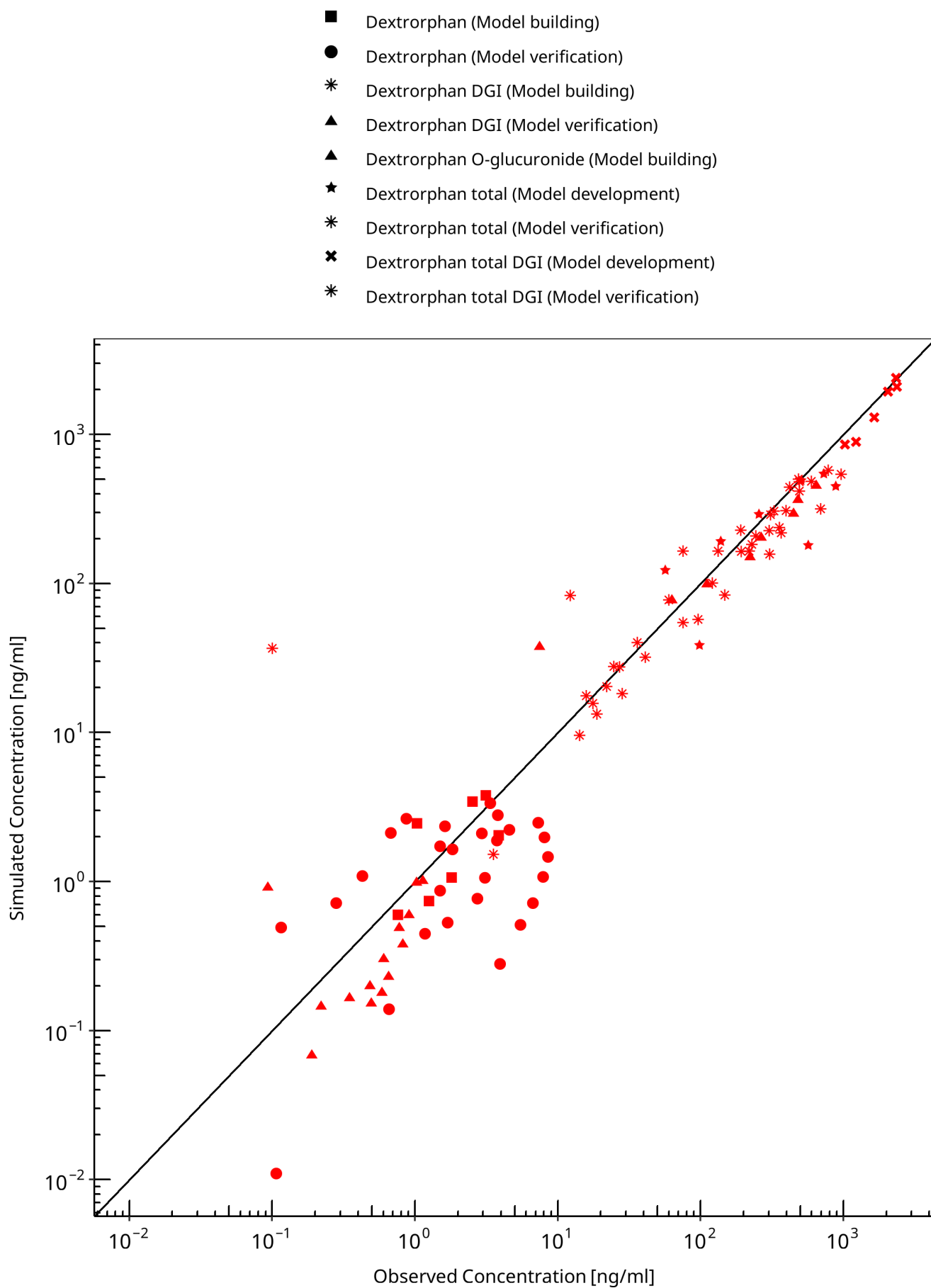
**Figure 3-1: Goodness of fit plot for Dextromethorphan concentration in plasma**



**Figure 3-2: Goodness of fit plot for Dextromethorphan concentration in plasma**

**Table 3-2: GMFE for Goodness of fit plot for Dextromethorphan metabolites concentration in plasma**

| Group                                      | GMFE |
|--------------------------------------------|------|
| Dextrorphan (Model building)               | 1.60 |
| Dextrorphan (Model verification)           | 3.09 |
| Dextrorphan DGI (Model building)           | 1.20 |
| Dextrorphan DGI (Model verification)       | 2.17 |
| Dextrorphan O-glucuronide (Model building) | 1.55 |
| Dextrorphan total (Model development)      | 1.72 |
| Dextrorphan total (Model verification)     | 1.79 |
| Dextrorphan total DGI (Model development)  | 1.18 |
| Dextrorphan total DGI (Model verification) | 1.53 |
| All                                        | 1.93 |



**Figure 3-3: Goodness of fit plot for Dextromethorphan metabolites concentration in plasma**

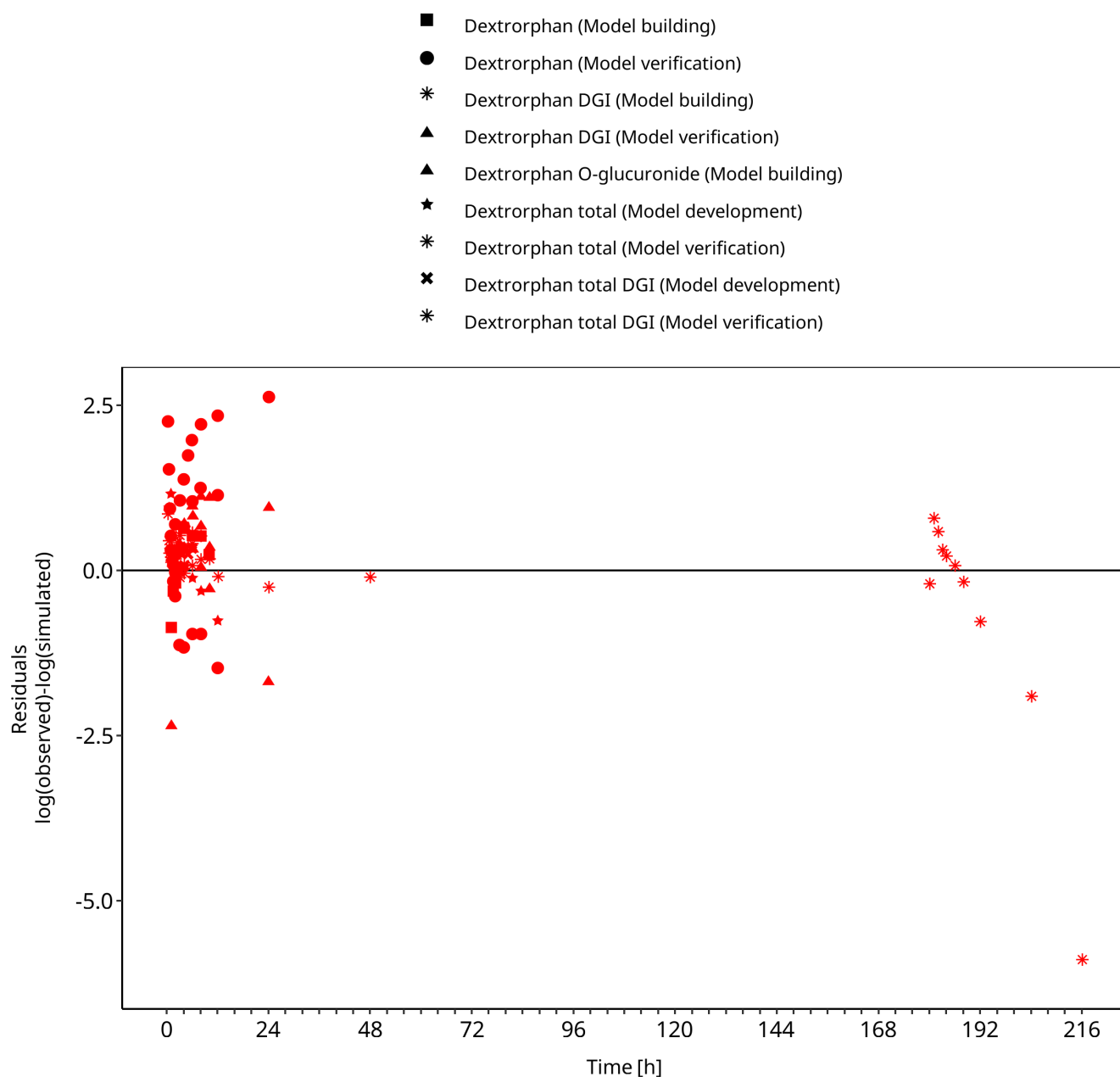


Figure 3-4: Goodness of fit plot for Dextromethorphan metabolites concentration in plasma

## 3.3 Concentration-Time Profiles

Simulated versus observed concentration-time profiles of all data listed in [Section 2.2.2](#) are presented below.

### 3.3.1 Model Building

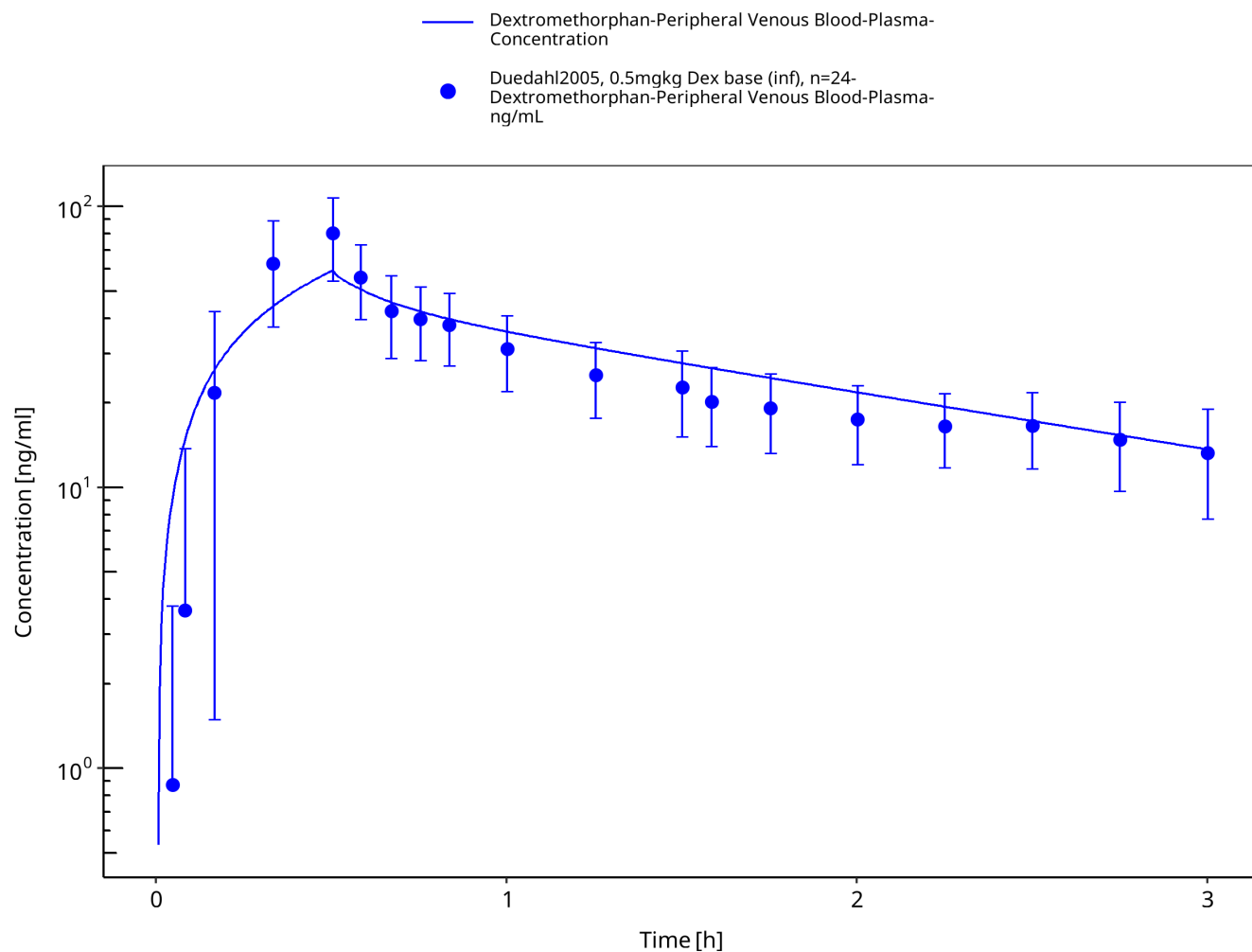


Figure 3-5: Duedahl 2005 EM, 0.5 mg/kg dextromethorphan base (infusion), n=24

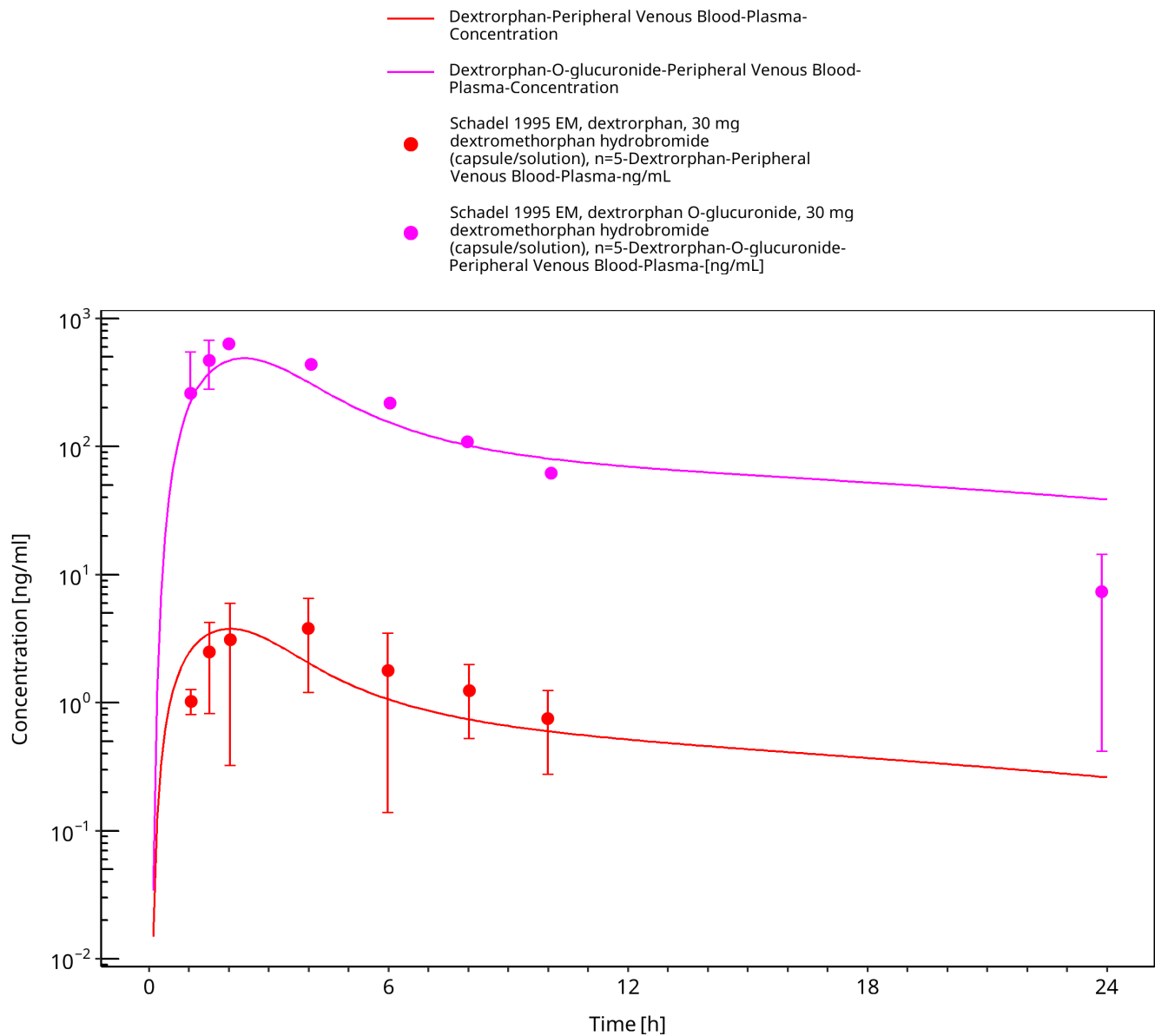


Figure 3-6: Schadel 1995 EM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=5

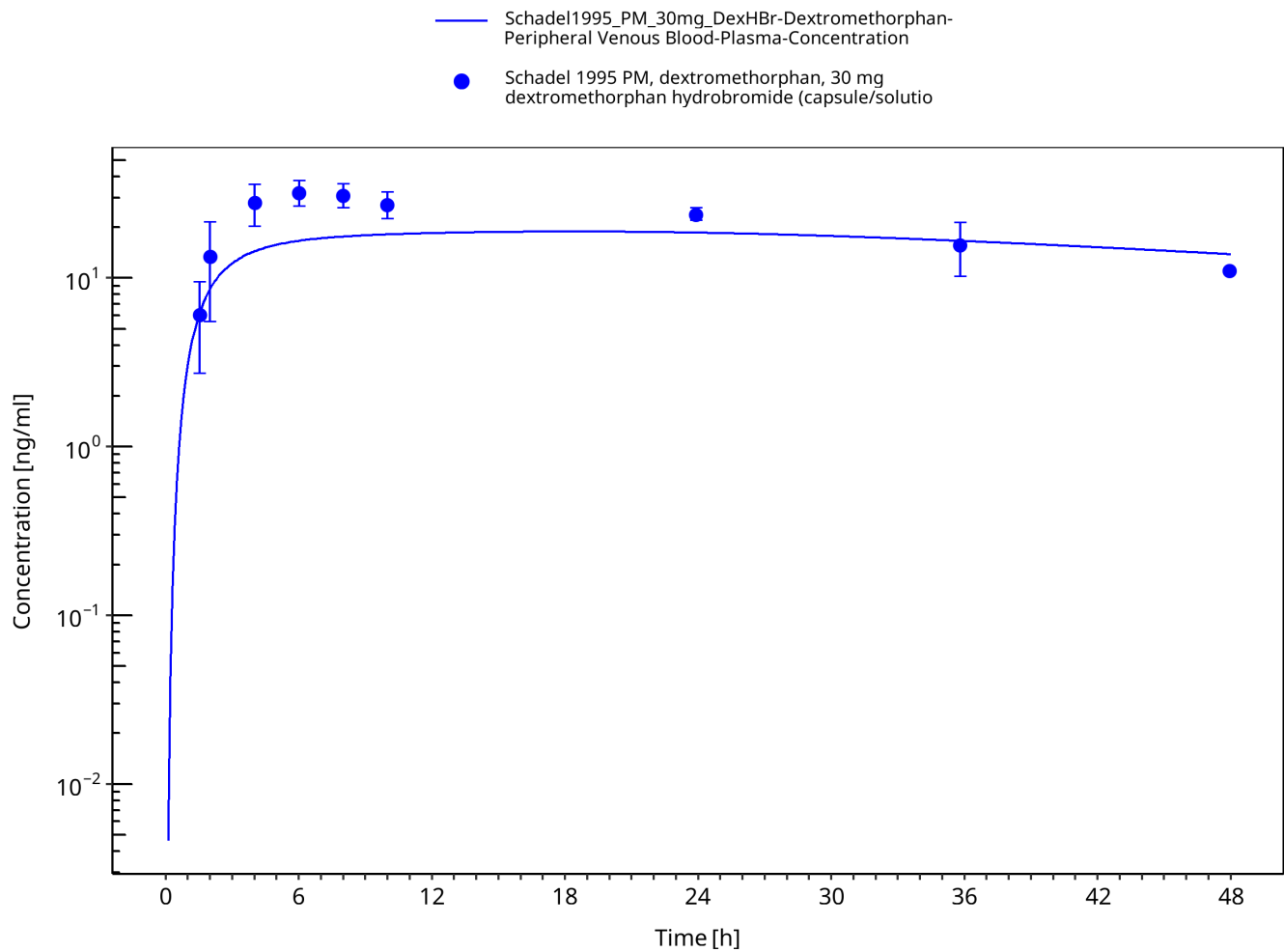


Figure 3-7: Schadel 1995 PM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=4

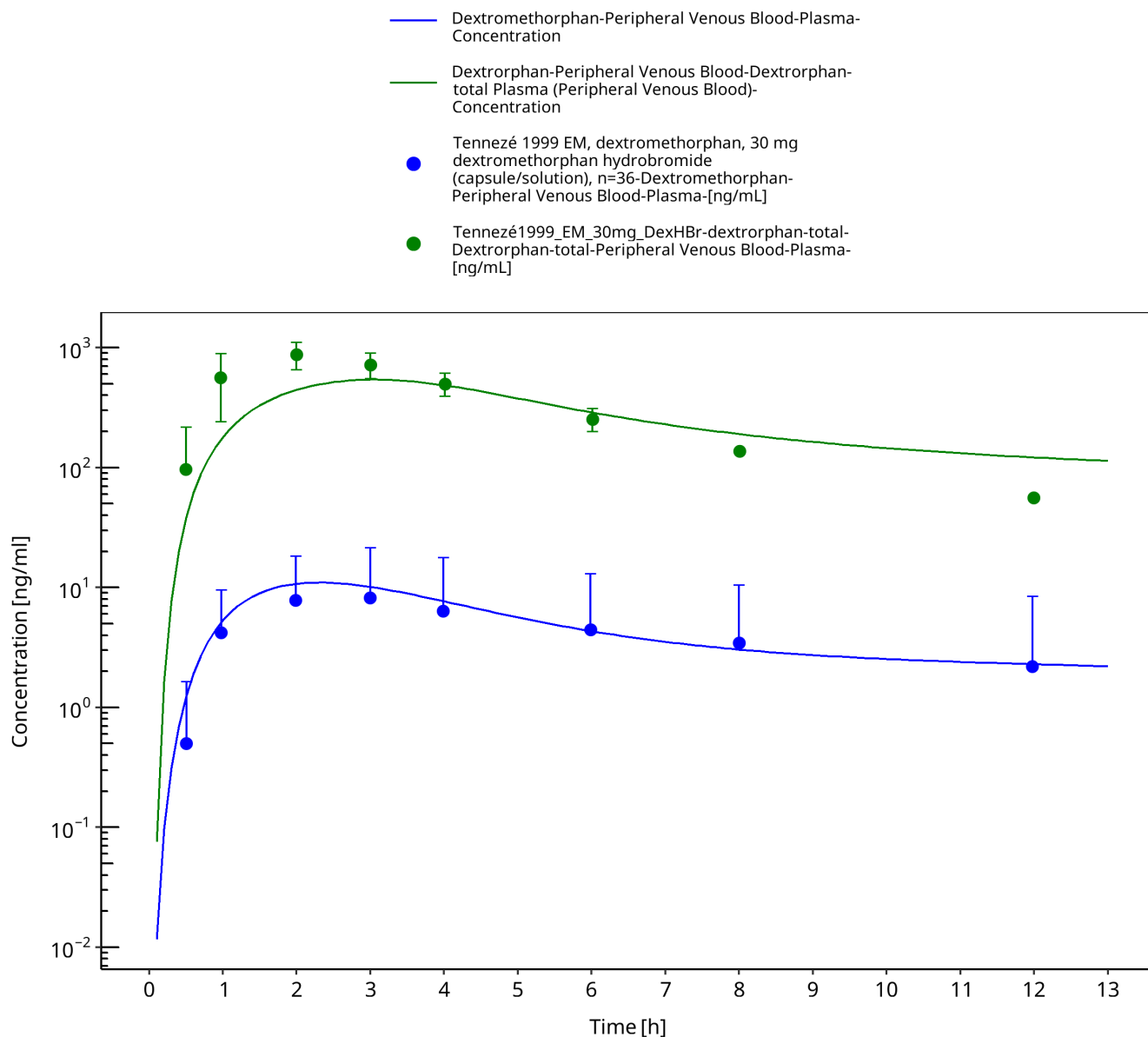


Figure 3-8: Tennezé 1999 EM, 80 mg dextromethorphan hydrobromide (capsule/solution), n=36

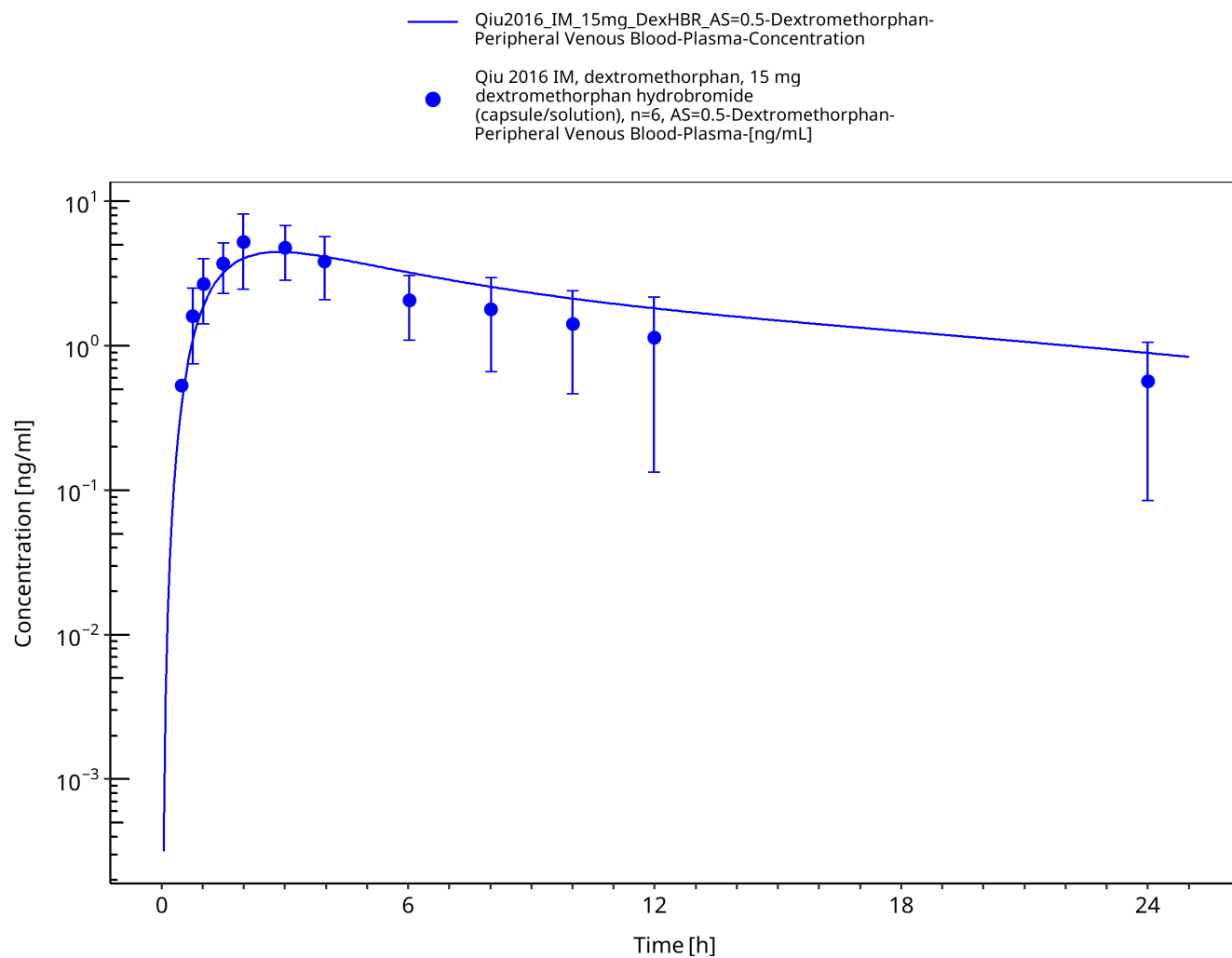


Figure 3-9: Qiu 2016 IM, 15 mg dextromethorphan hydrobromide (capsule/solution), n=6, AS=0.5

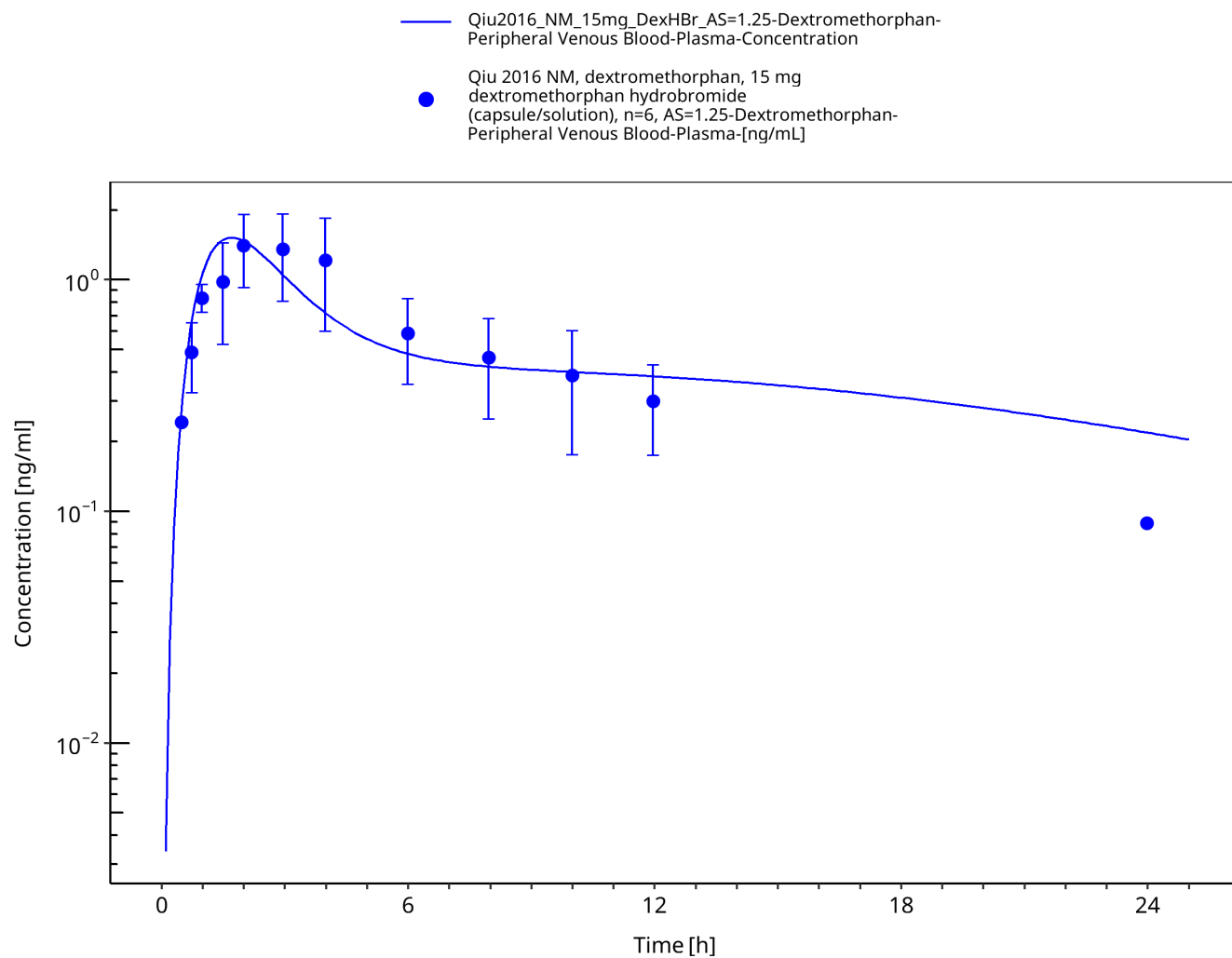


Figure 3-10: Qiu 2016 NM, 15 mg dextromethorphan hydrobromide (capsule/solution), n=6, AS=1.25

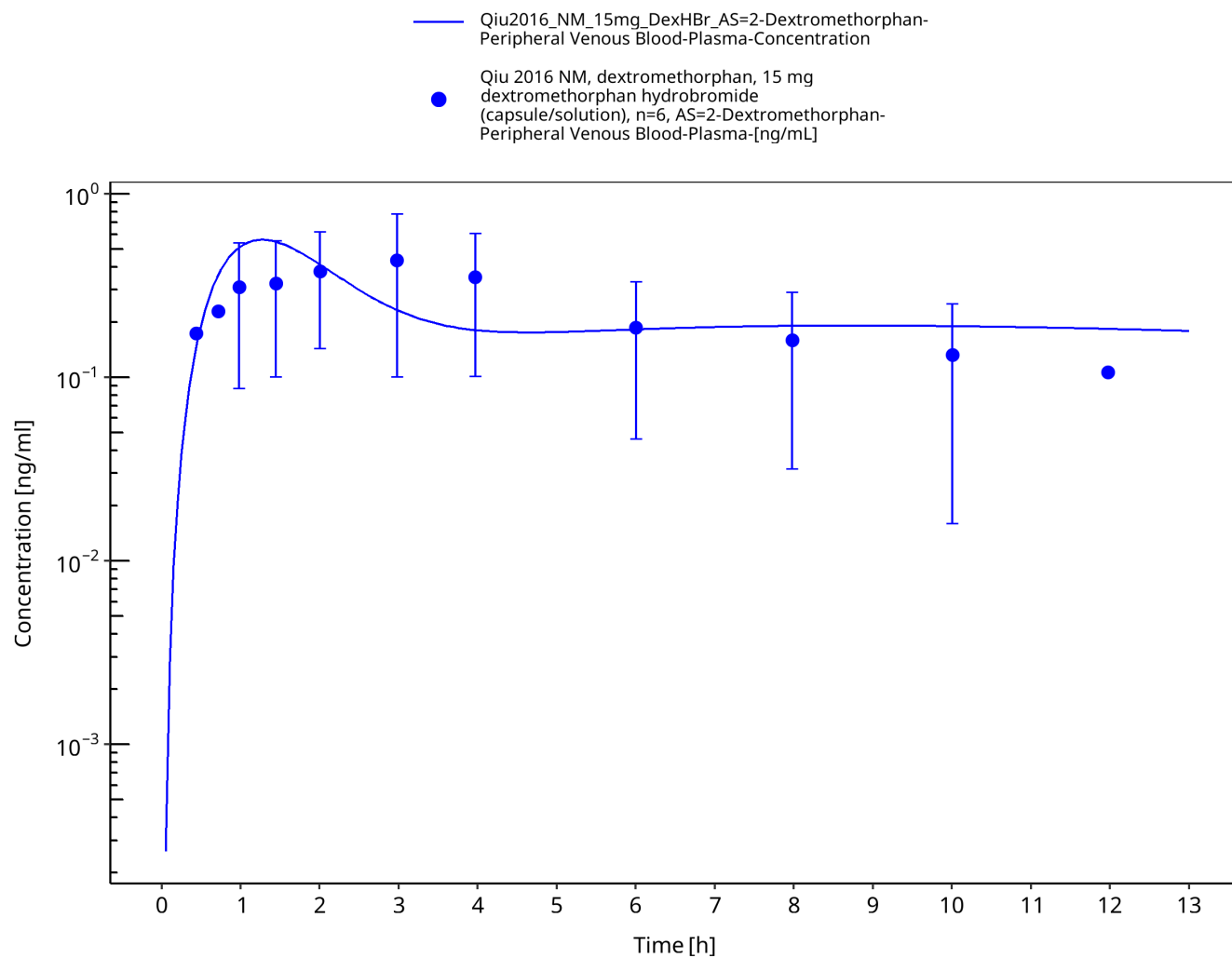


Figure 3-11: Qiu 2016 NM, 15 mg dextromethorphan hydrobromide (capsule/solution), n=6, AS=2

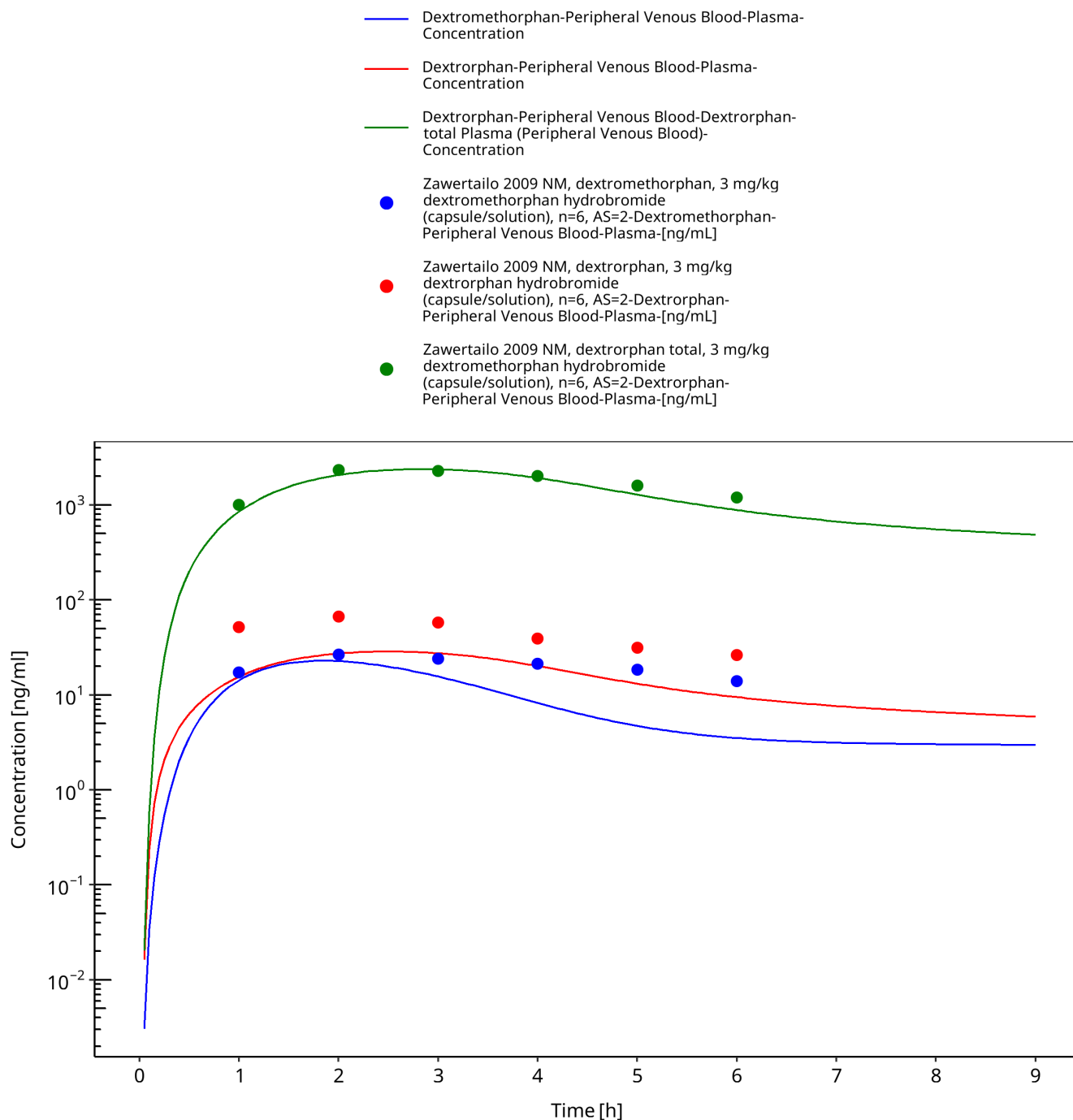
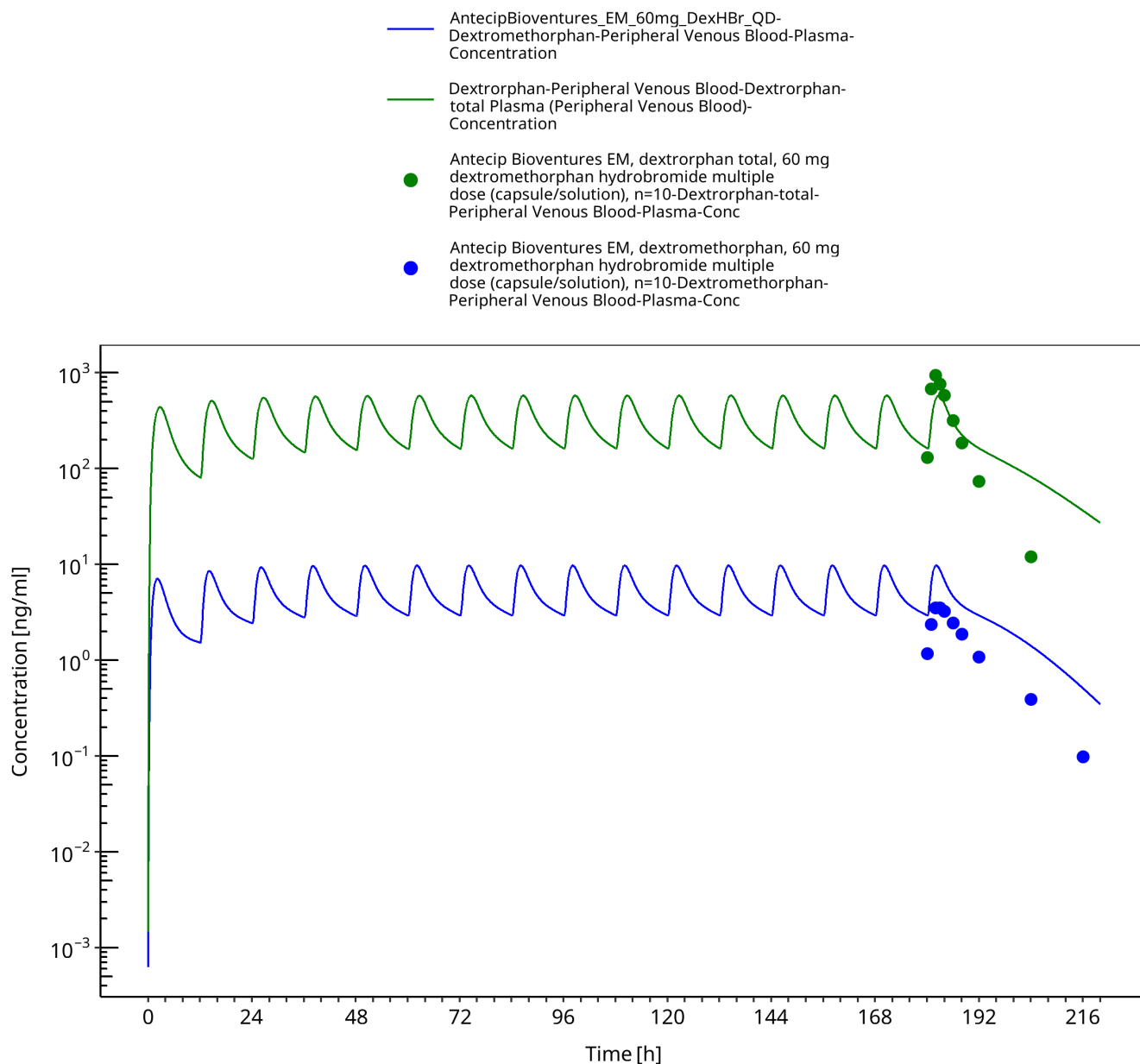


Figure 3-12: Zawertailo 2009 NM, 3 mg/kg dextromethorphan hydrobromide (capsule/solution), n=6, AS=2

### 3.3.2 Model Verification



**Figure 3-13: Antecip Bioventures EM, 60 mg dextromethorphan hydrobromide multiple dose (capsule/solution), n=10**

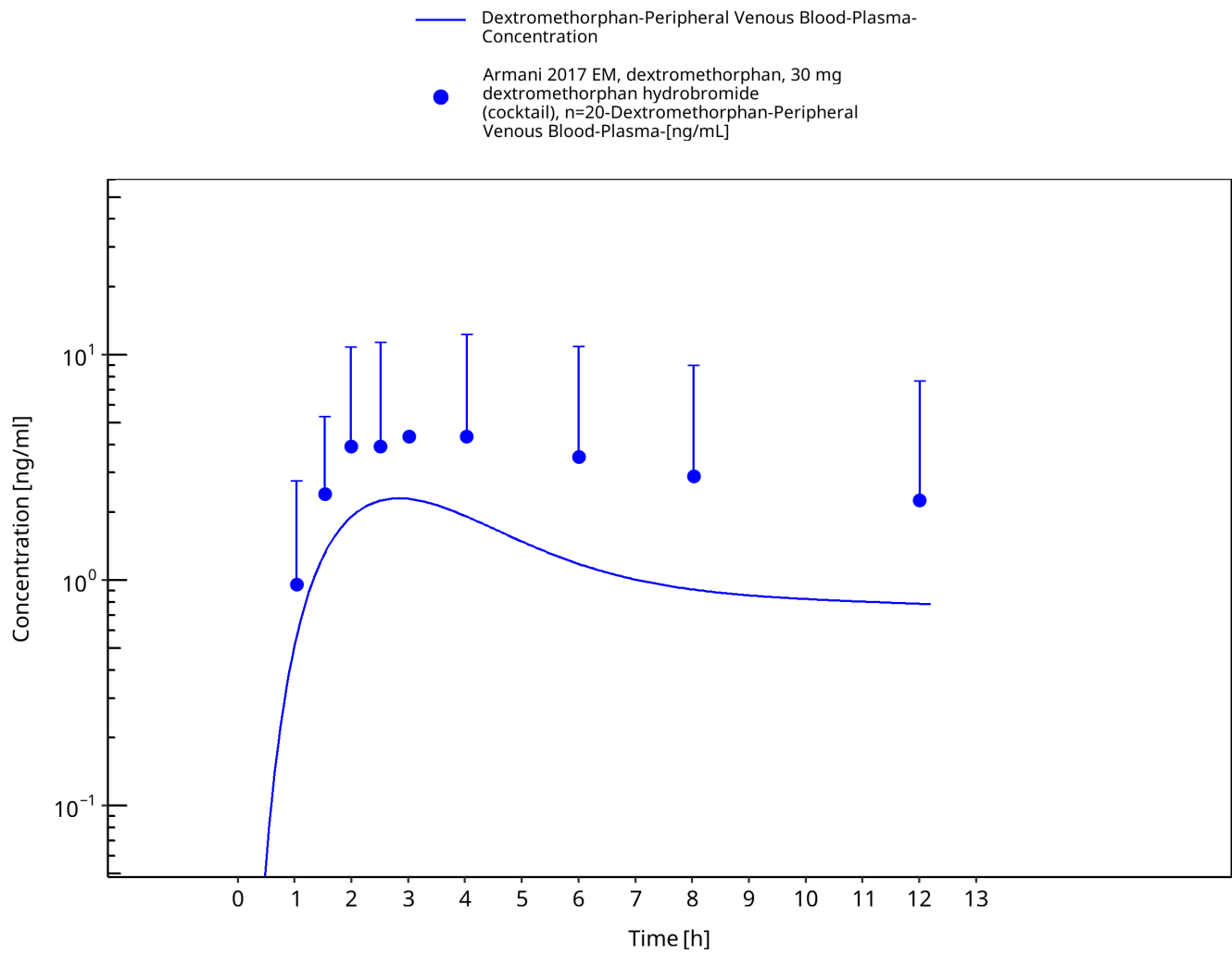
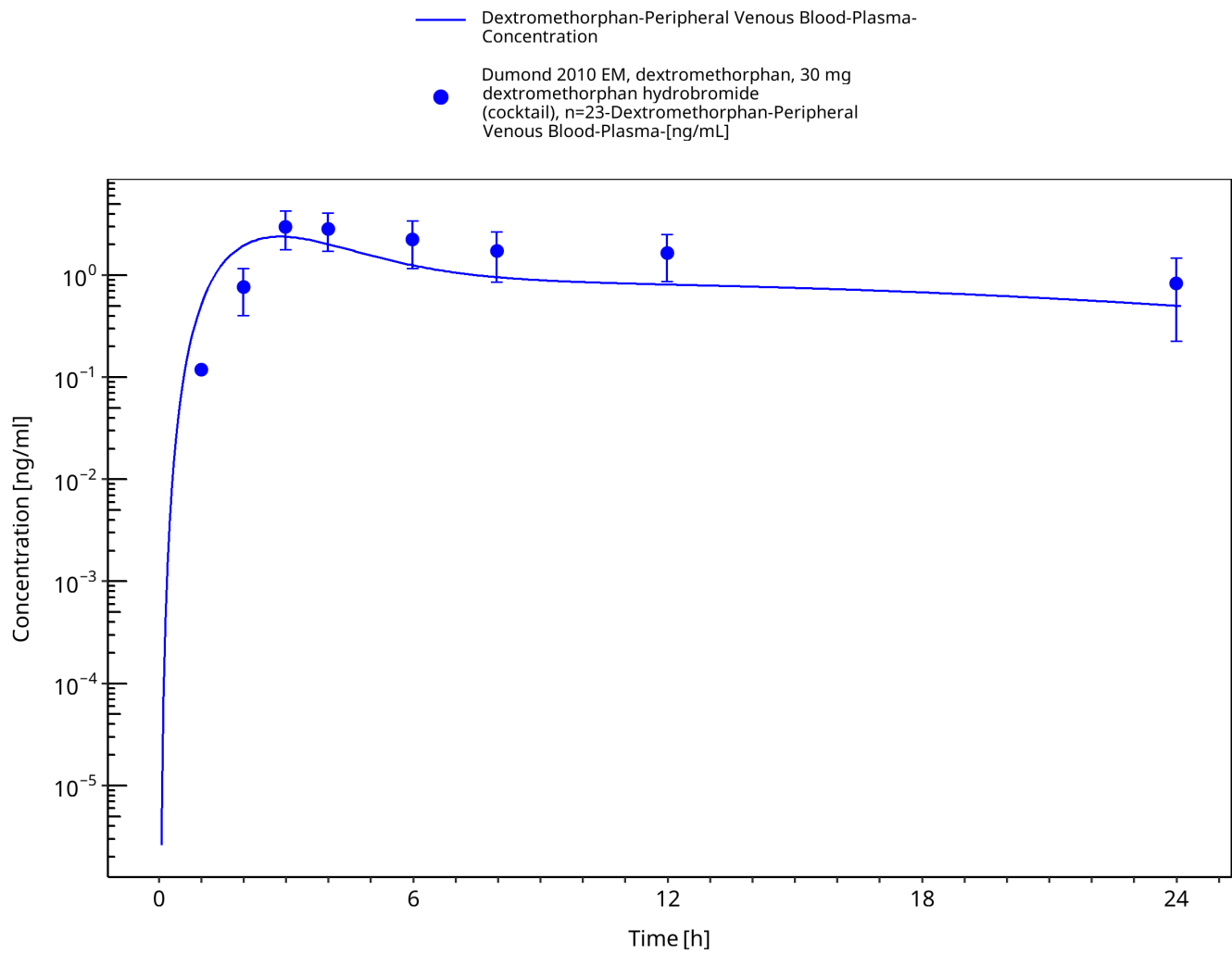


Figure 3-14: Armani 2017 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=20



**Figure 3-15: Dumond 2010 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=23**

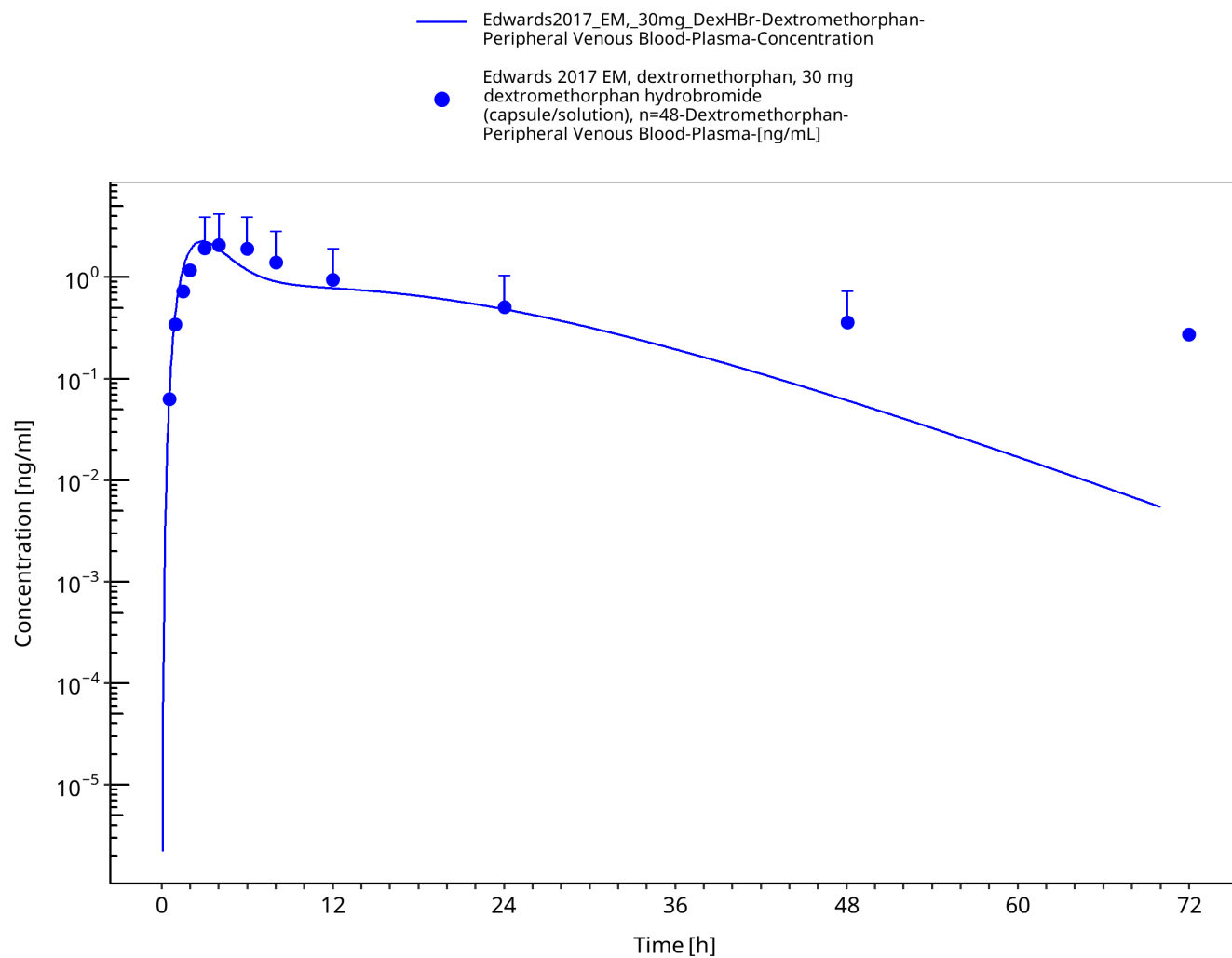


Figure 3-16: Edwards 2017 EM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=48

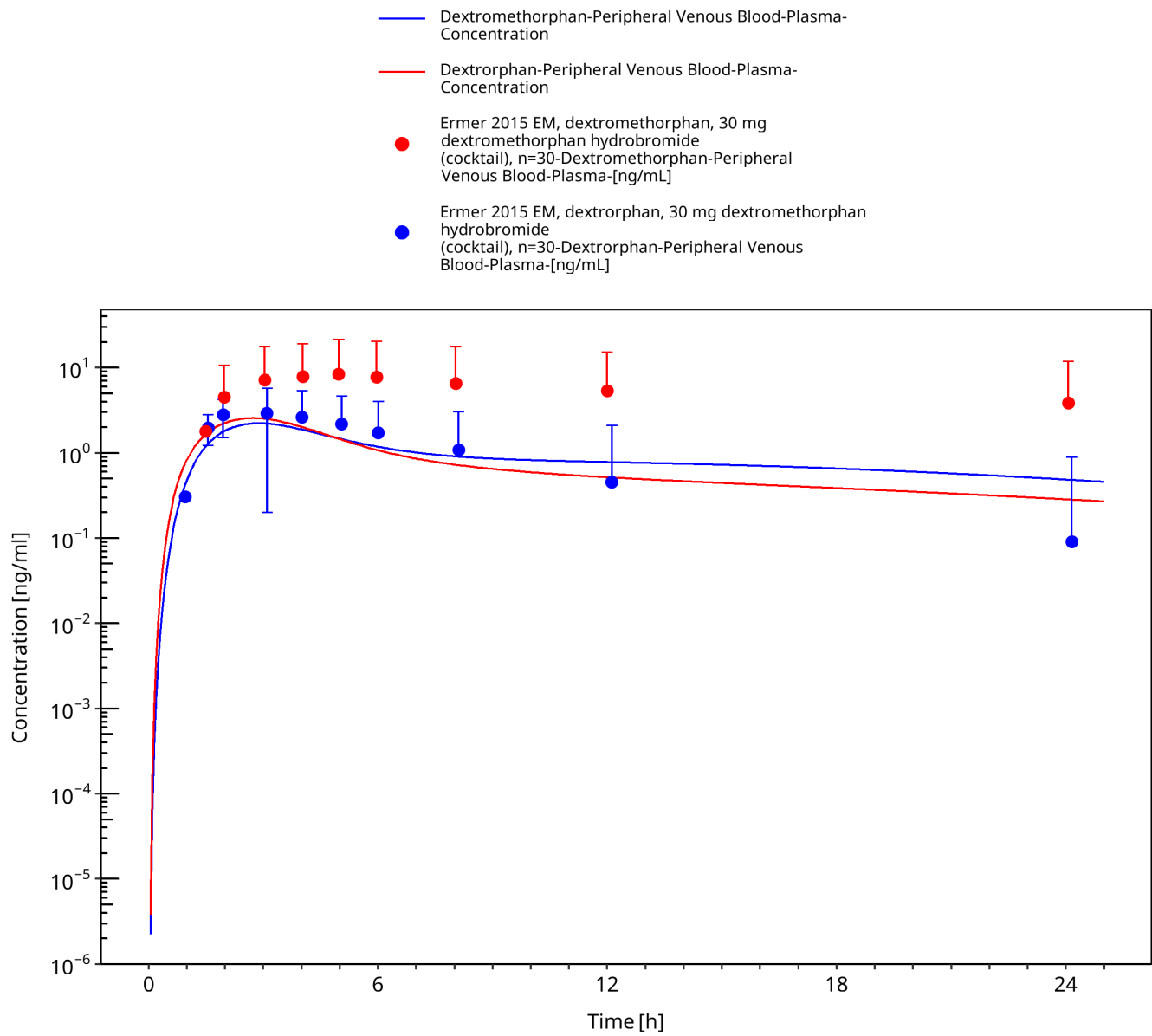


Figure 3-17: Ermer 2015 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=30

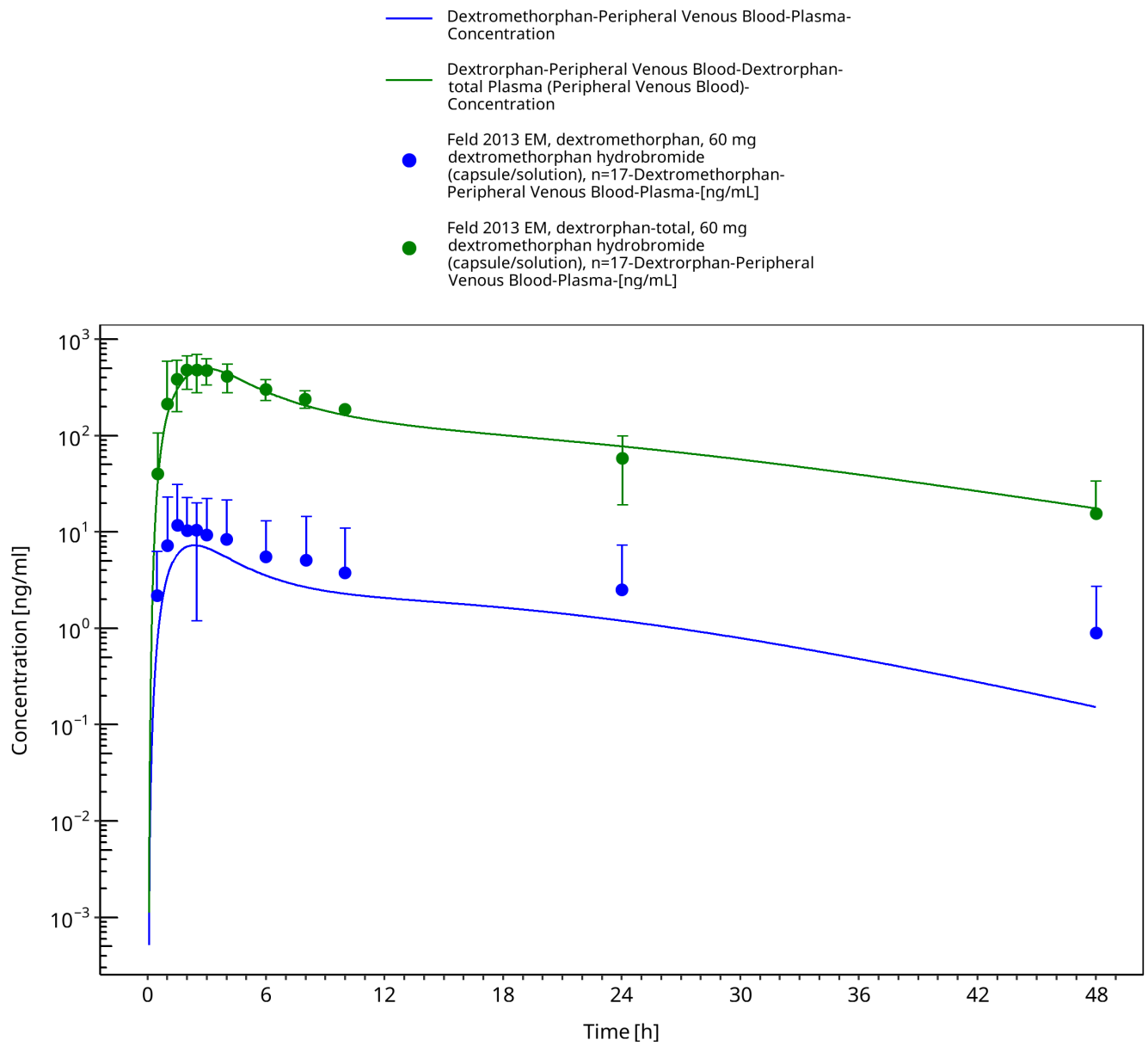
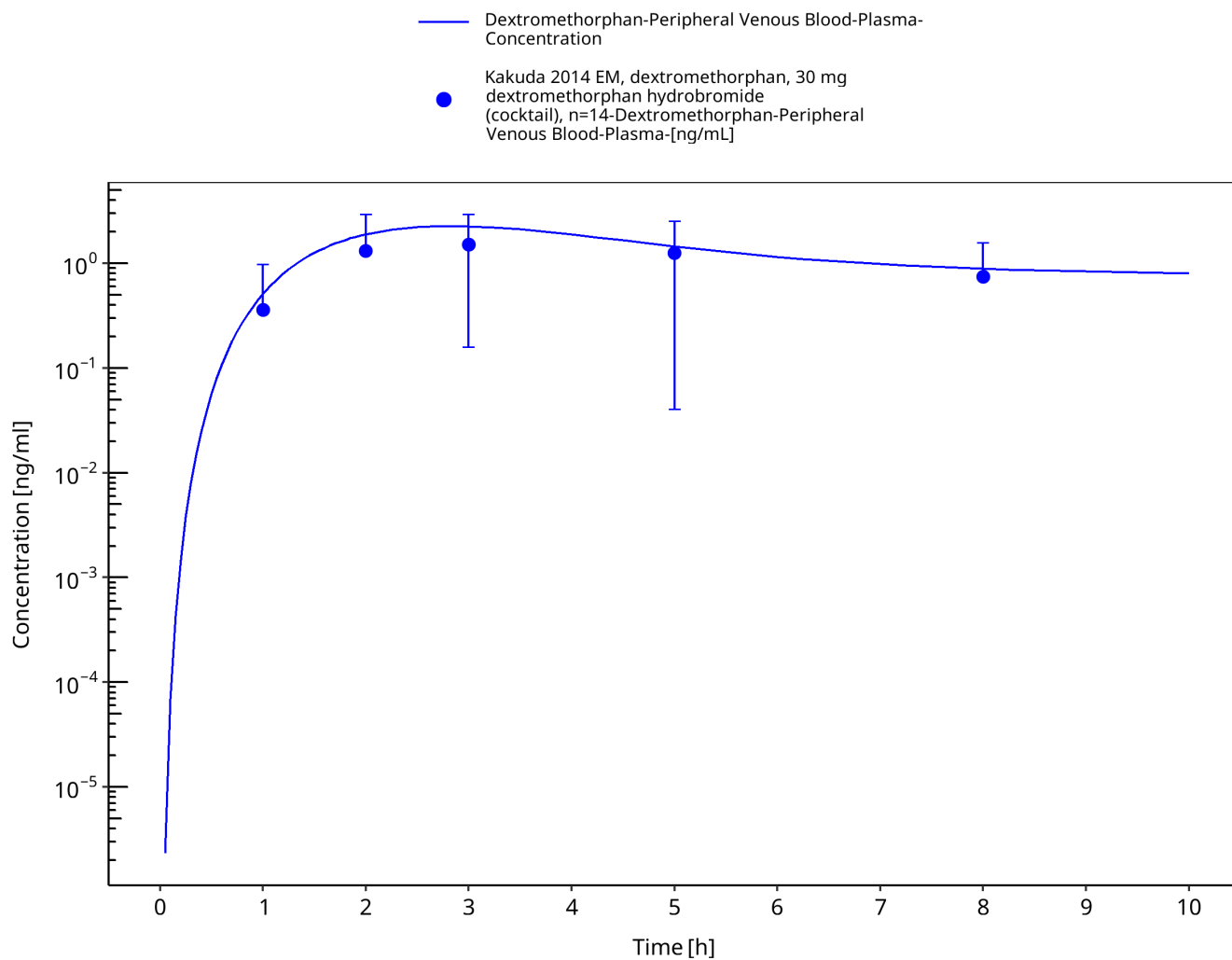
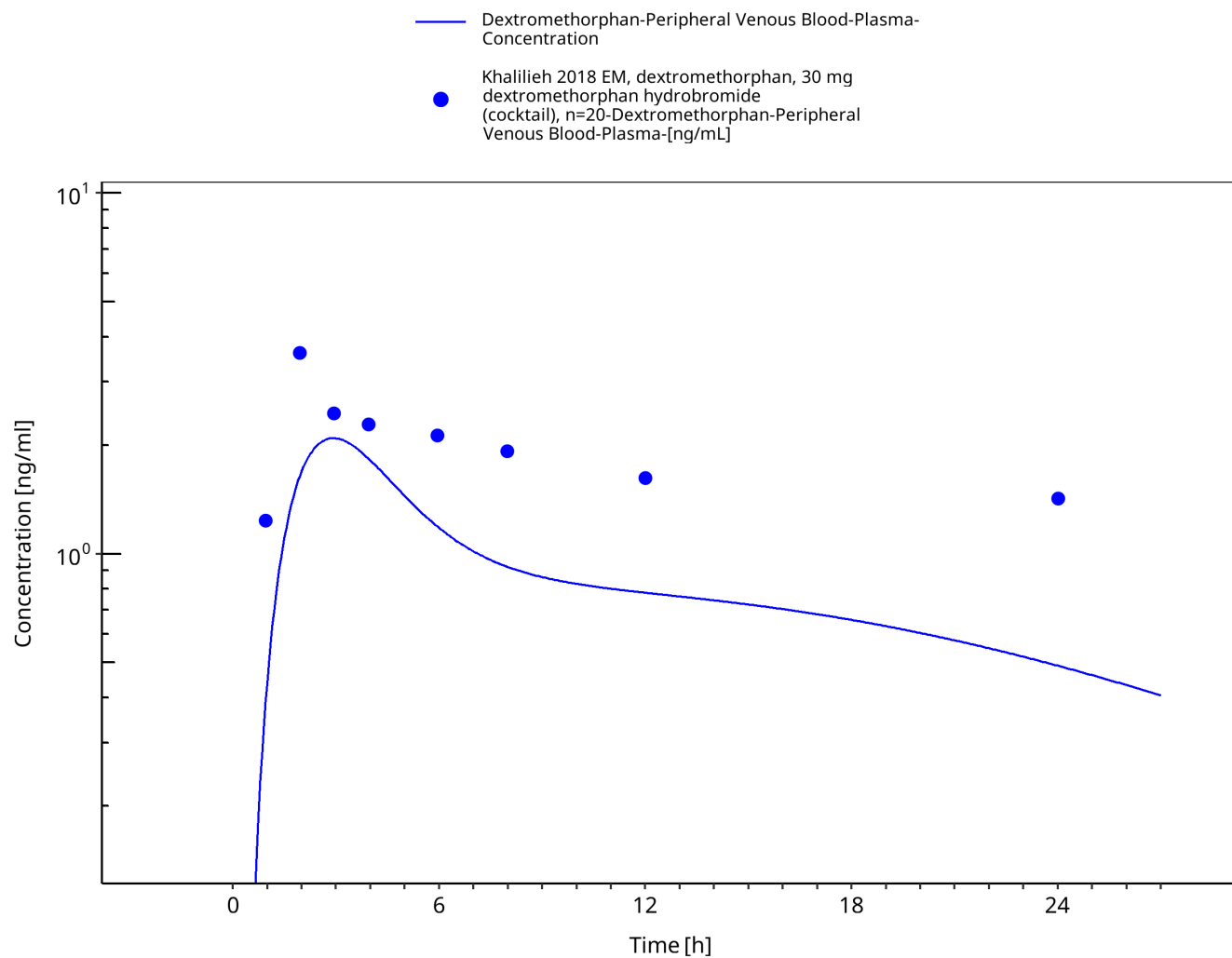


Figure 3-18: Feld 2013 EM, 60 mg dextromethorphan hydrobromide (capsule/solution), n=17



**Figure 3-19: Kakuda 2014 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=14**



**Figure 3-20: Khalilieh 2018 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=20**

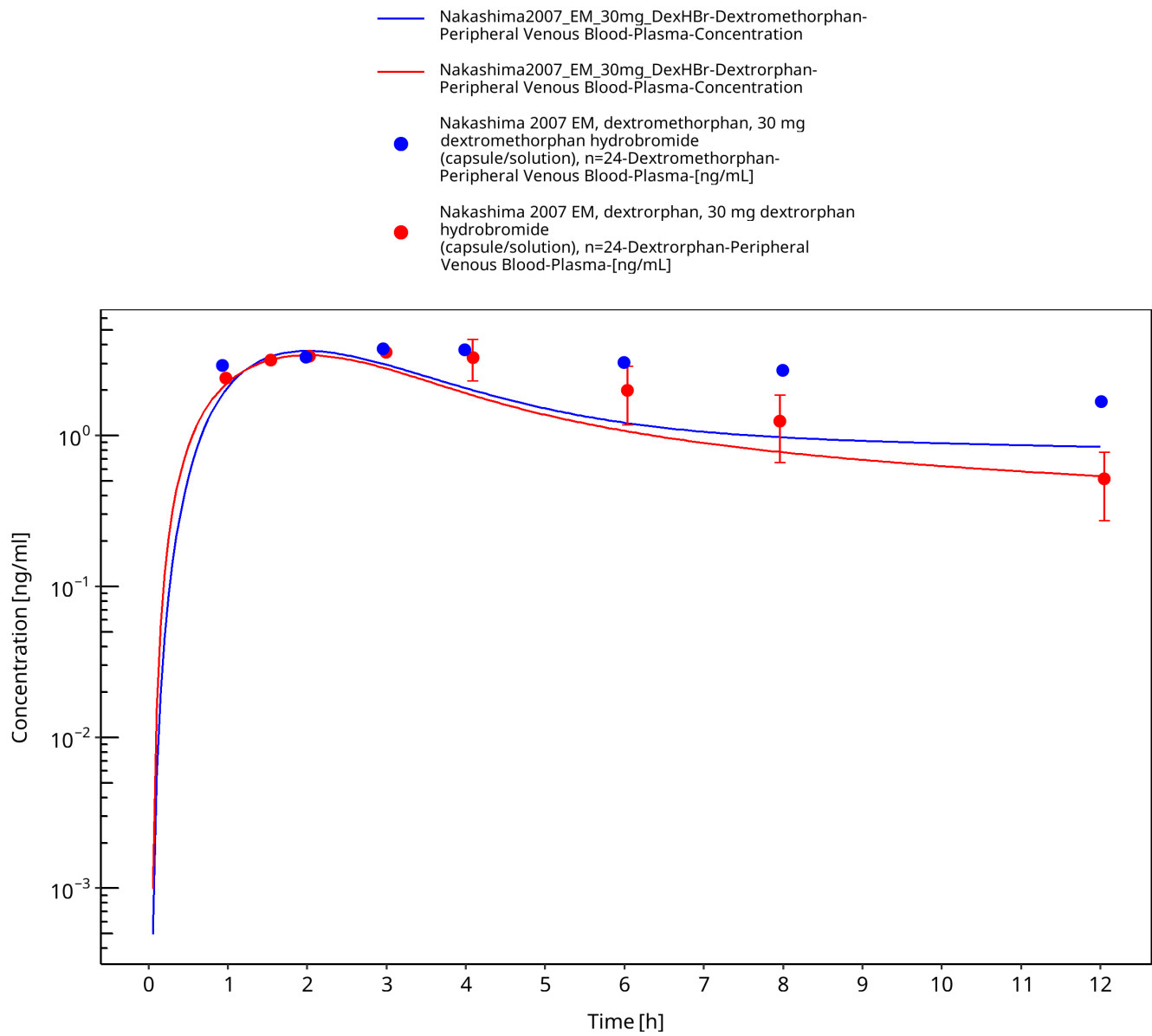


Figure 3-21: Nakashima 2007 EM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=24

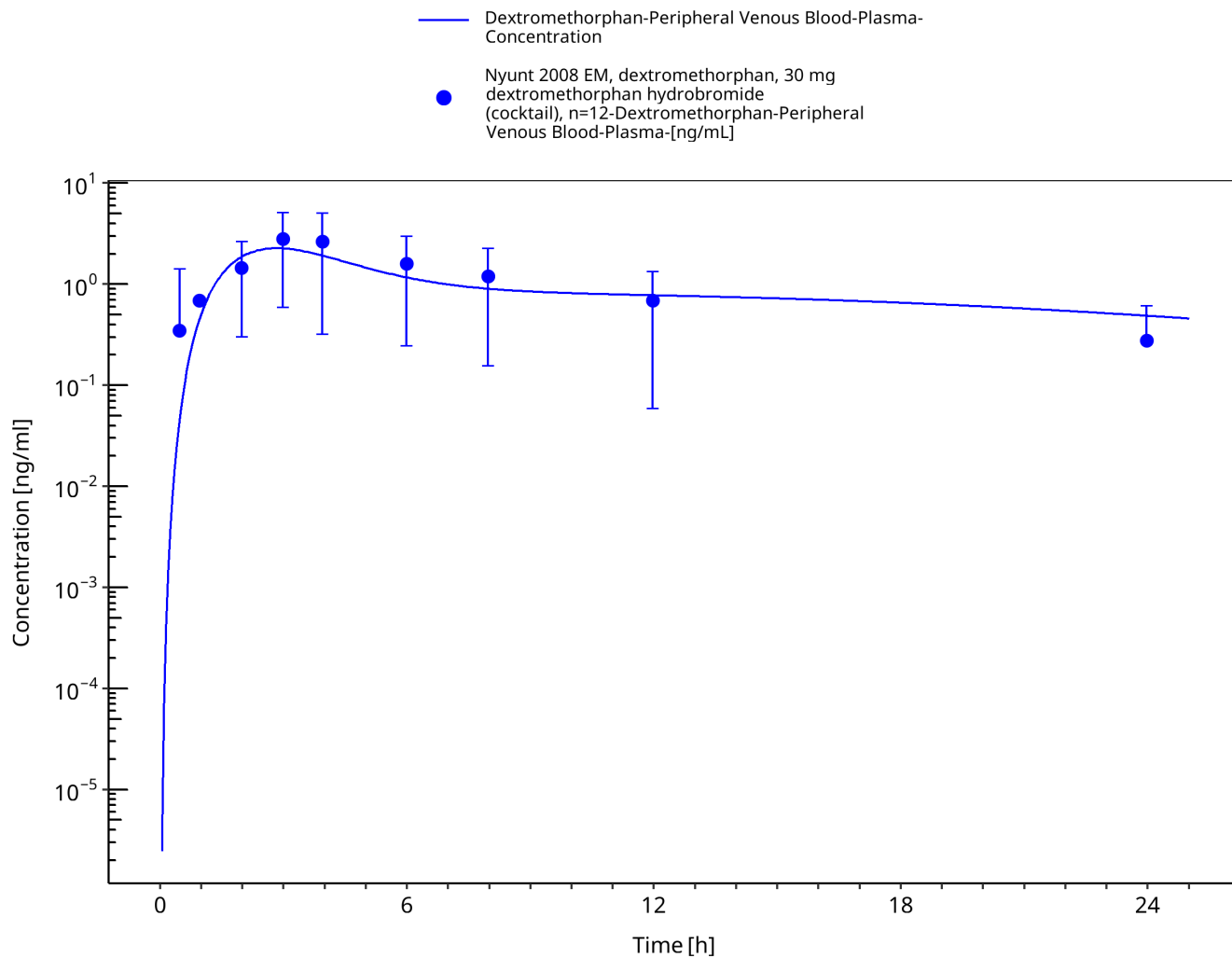


Figure 3-22: Nyunt 2008 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=12

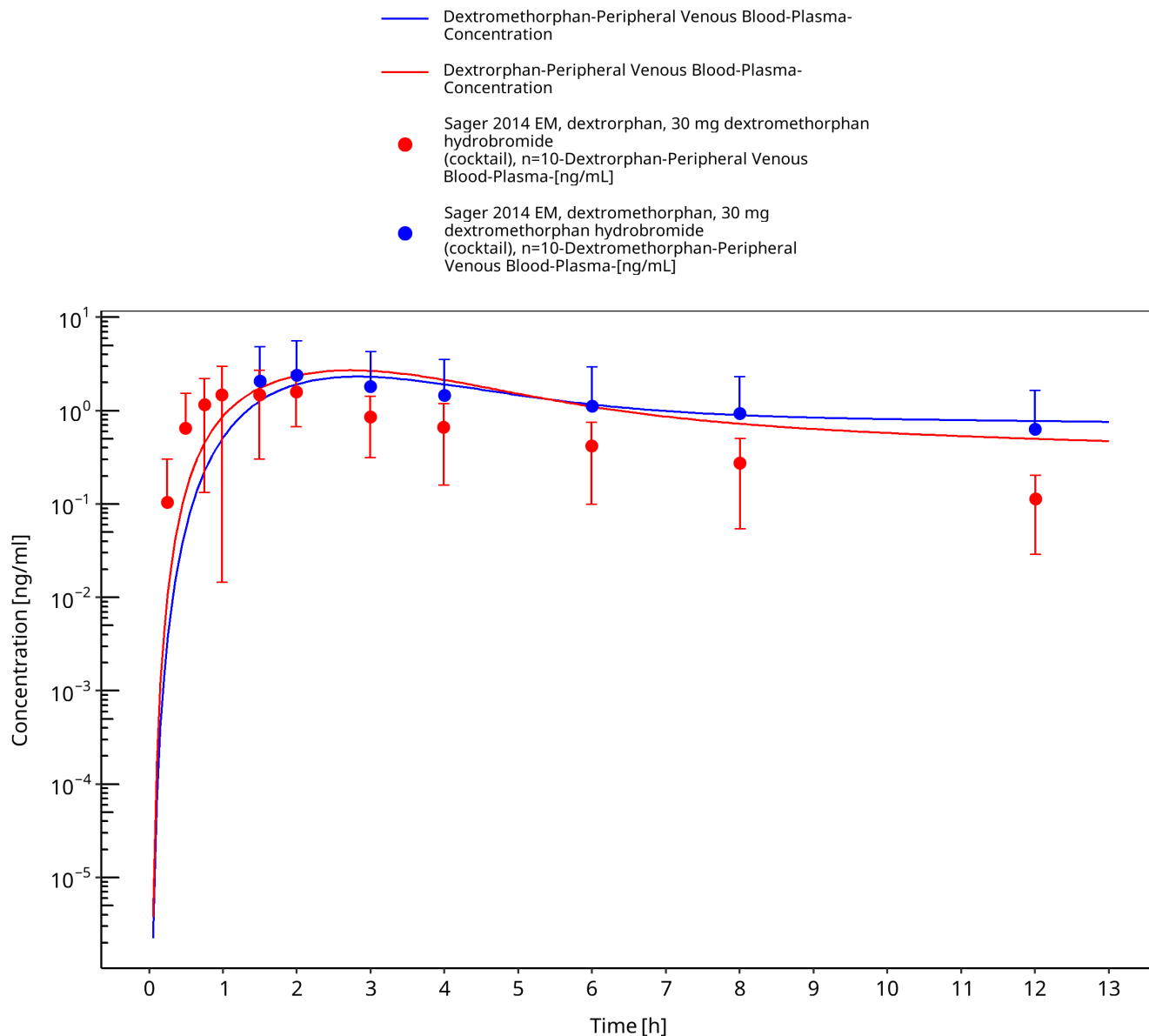


Figure 3-23: Sager 2014 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=10

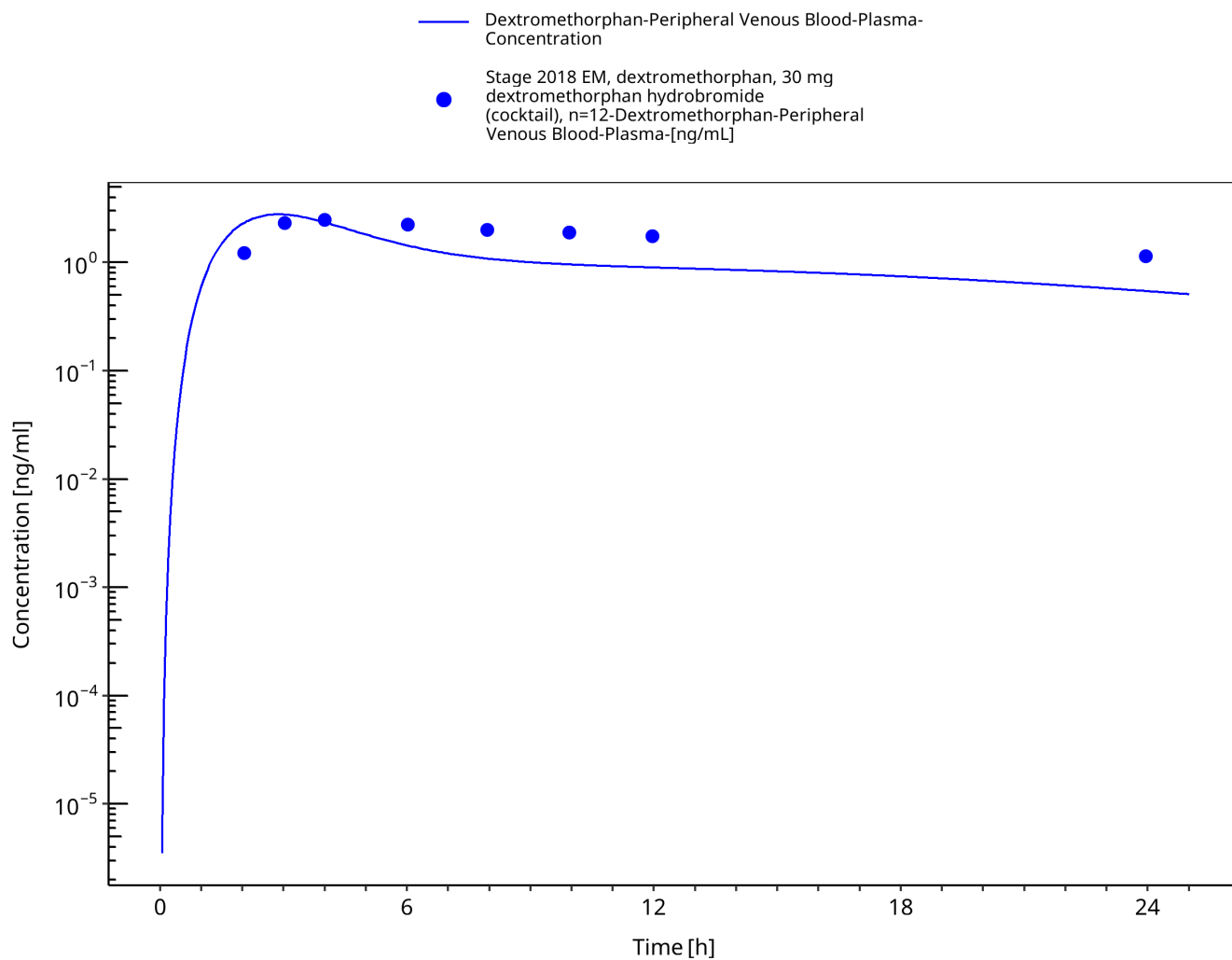


Figure 3-24: Stage 2018 EM, 30 mg dextromethorphan hydrobromide (cocktail), n=12

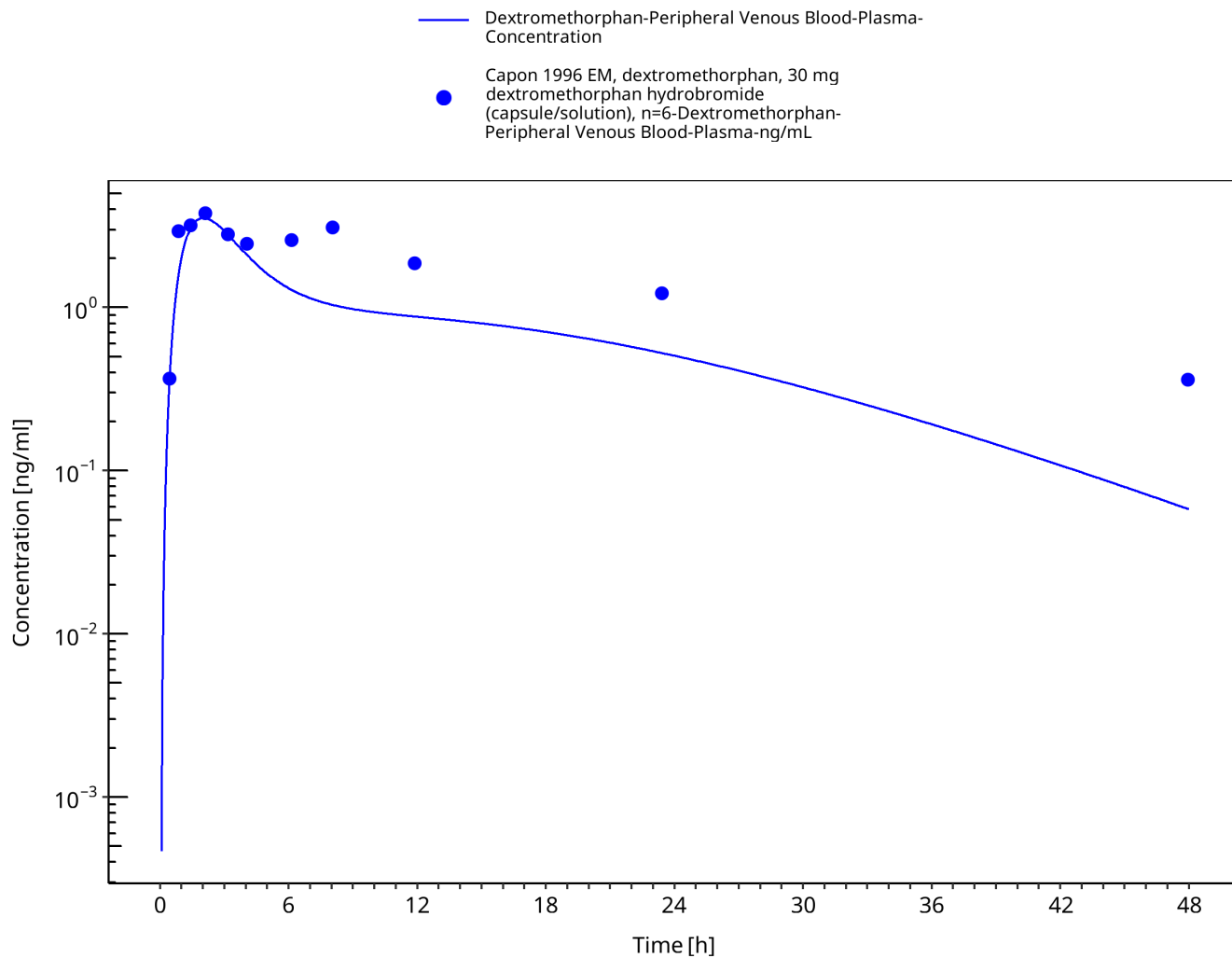


Figure 3-25: Capon 1996 EM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=6

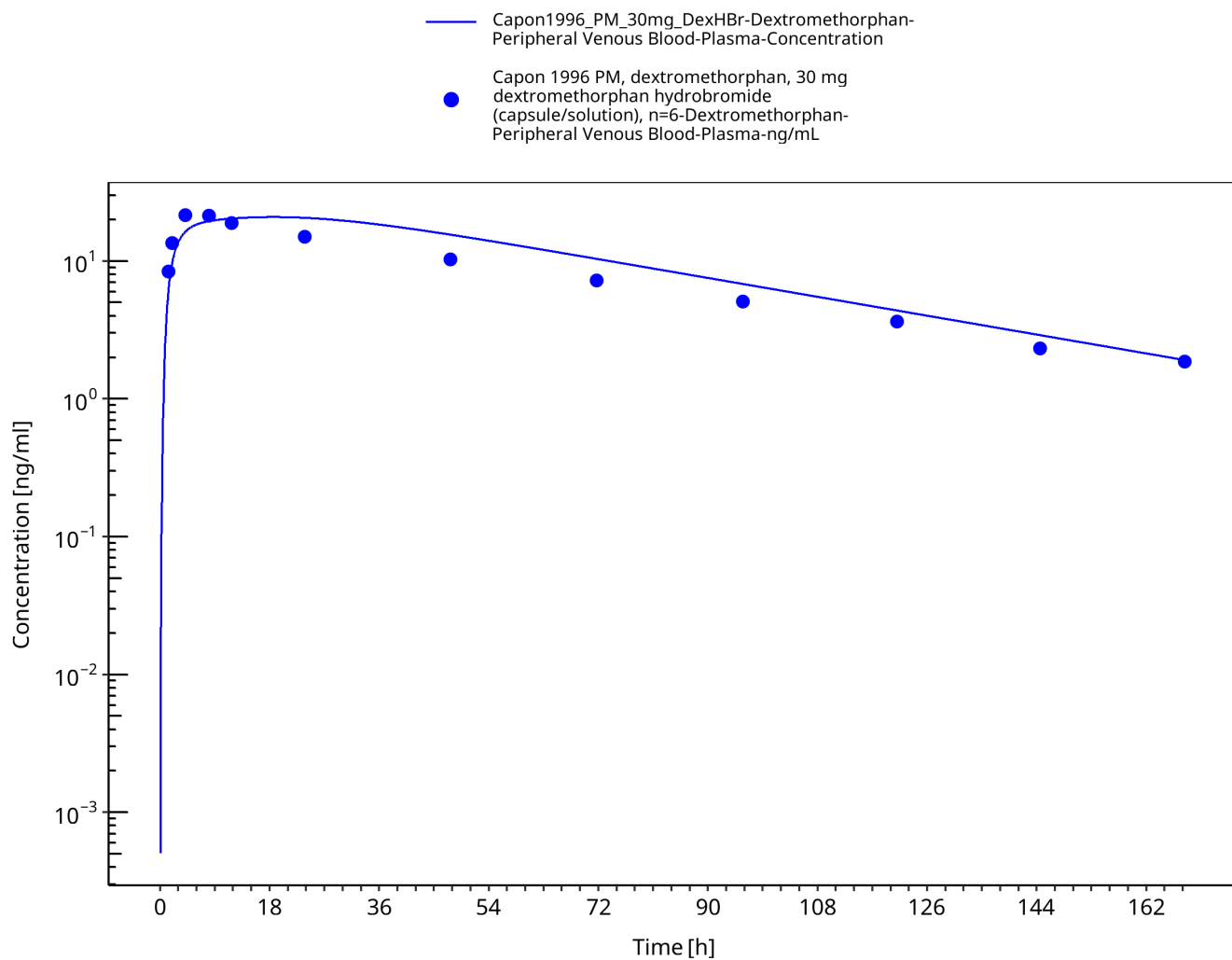


Figure 3-26: Capon 1996 PM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=6

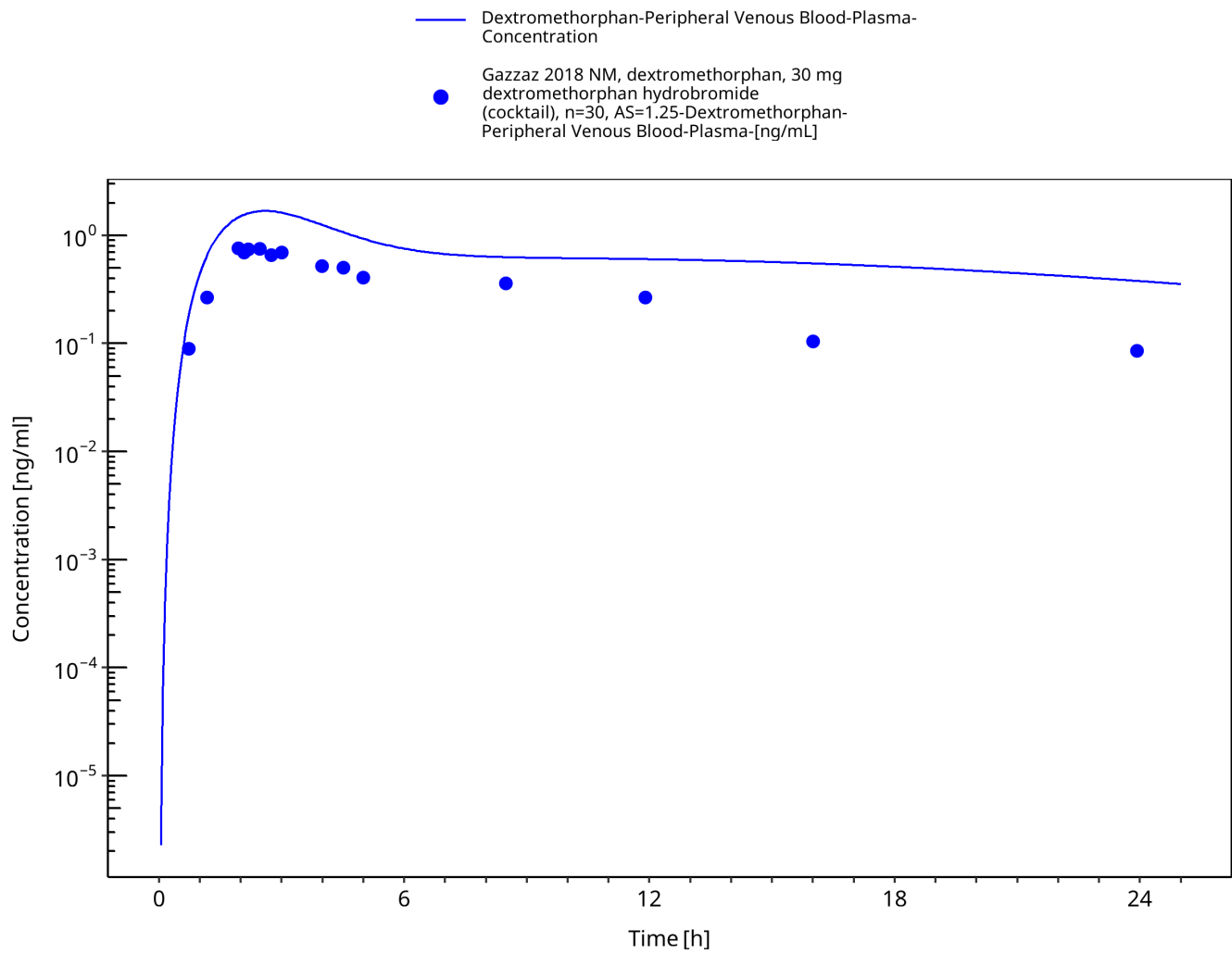


Figure 3-27: Gazzaz 2018 NM, 30 mg dextromethorphan hydrobromide (cocktail), n=30, AS=1.25

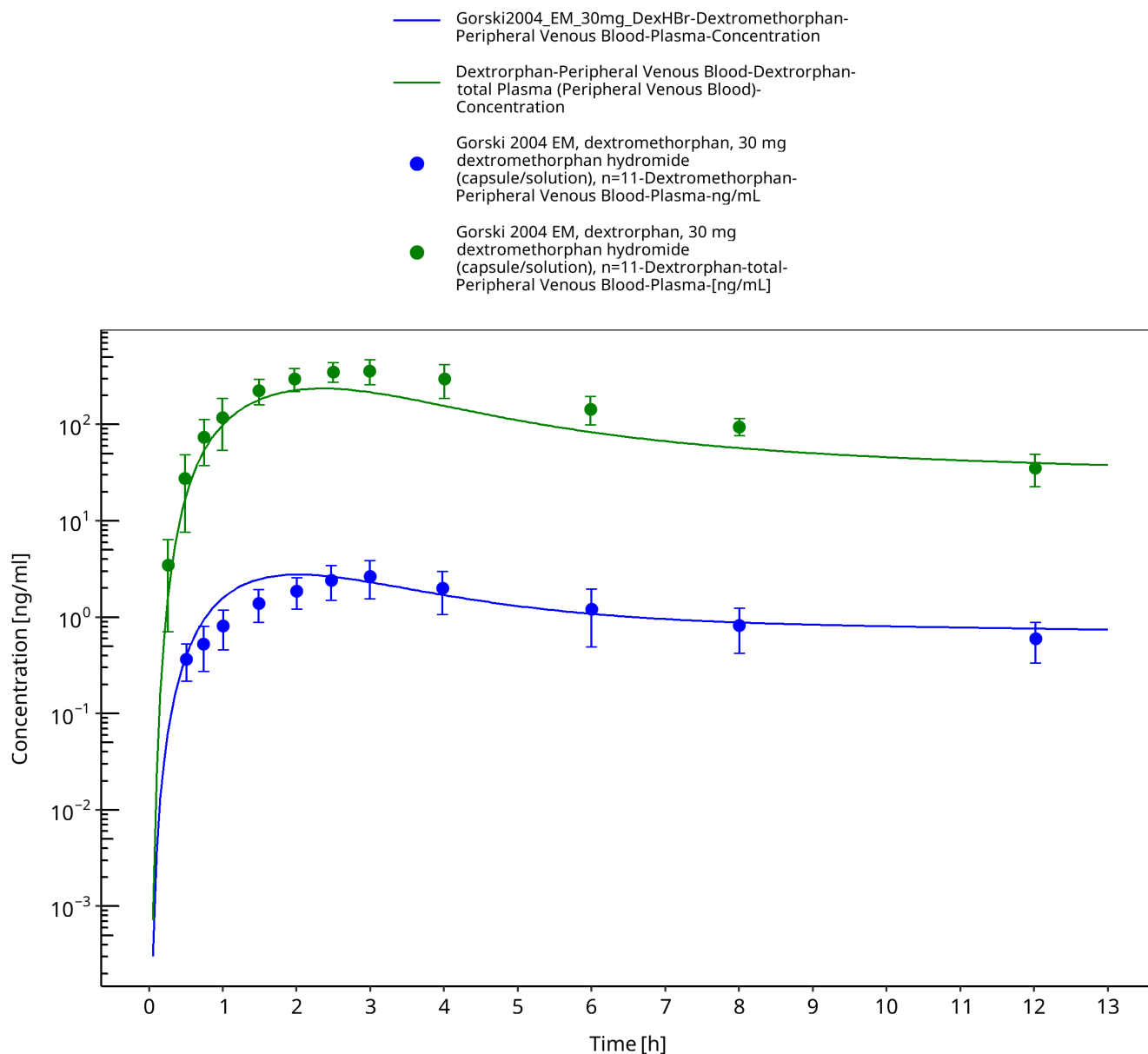


Figure 3-28: Gorski 2004 EM, 30 mg dextromethorphan hydromide (capsule/solution), n=11

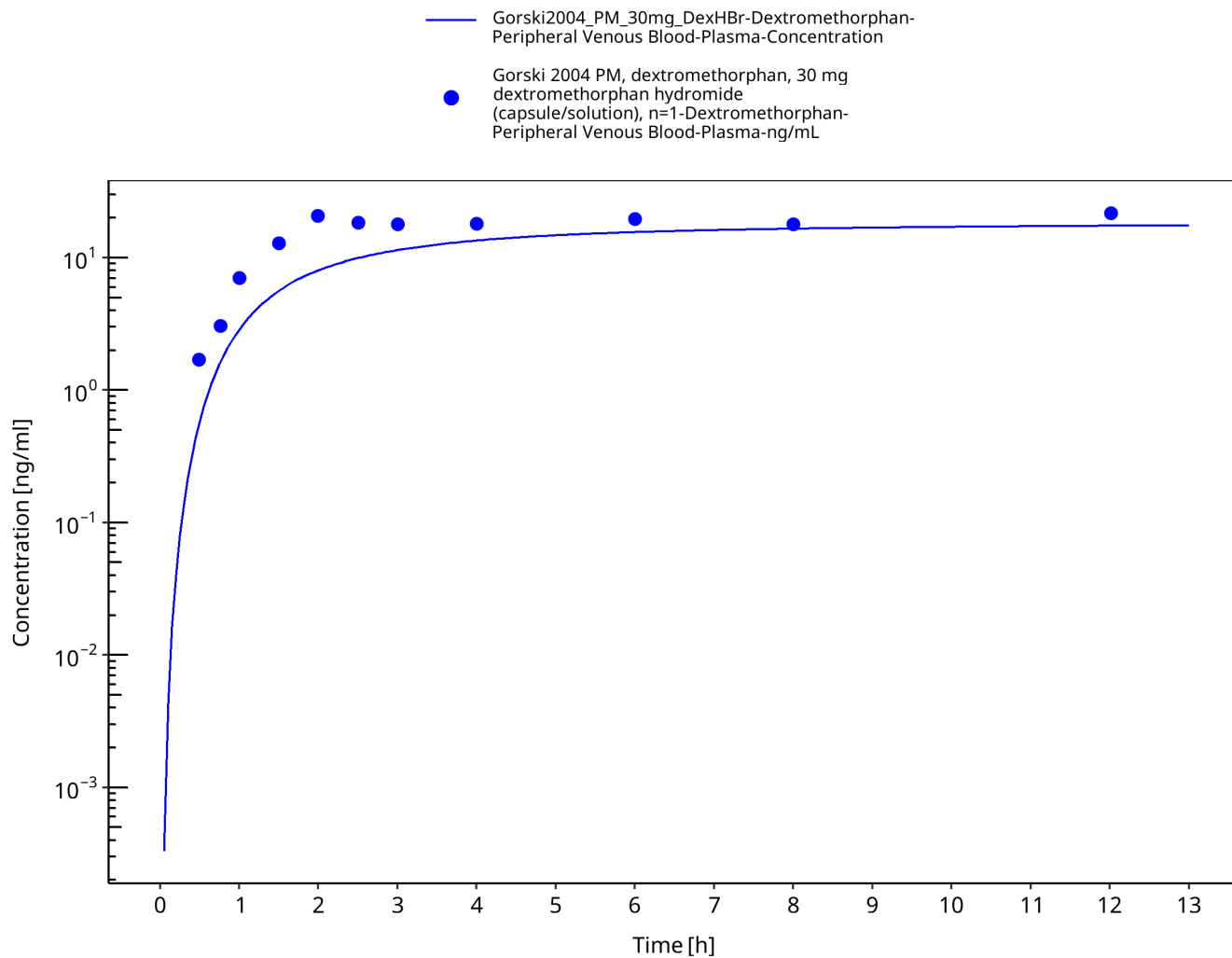


Figure 3-29: Gorski 2004 PM, 30 mg dextromethorphan hydromide (capsule/solution), n=1

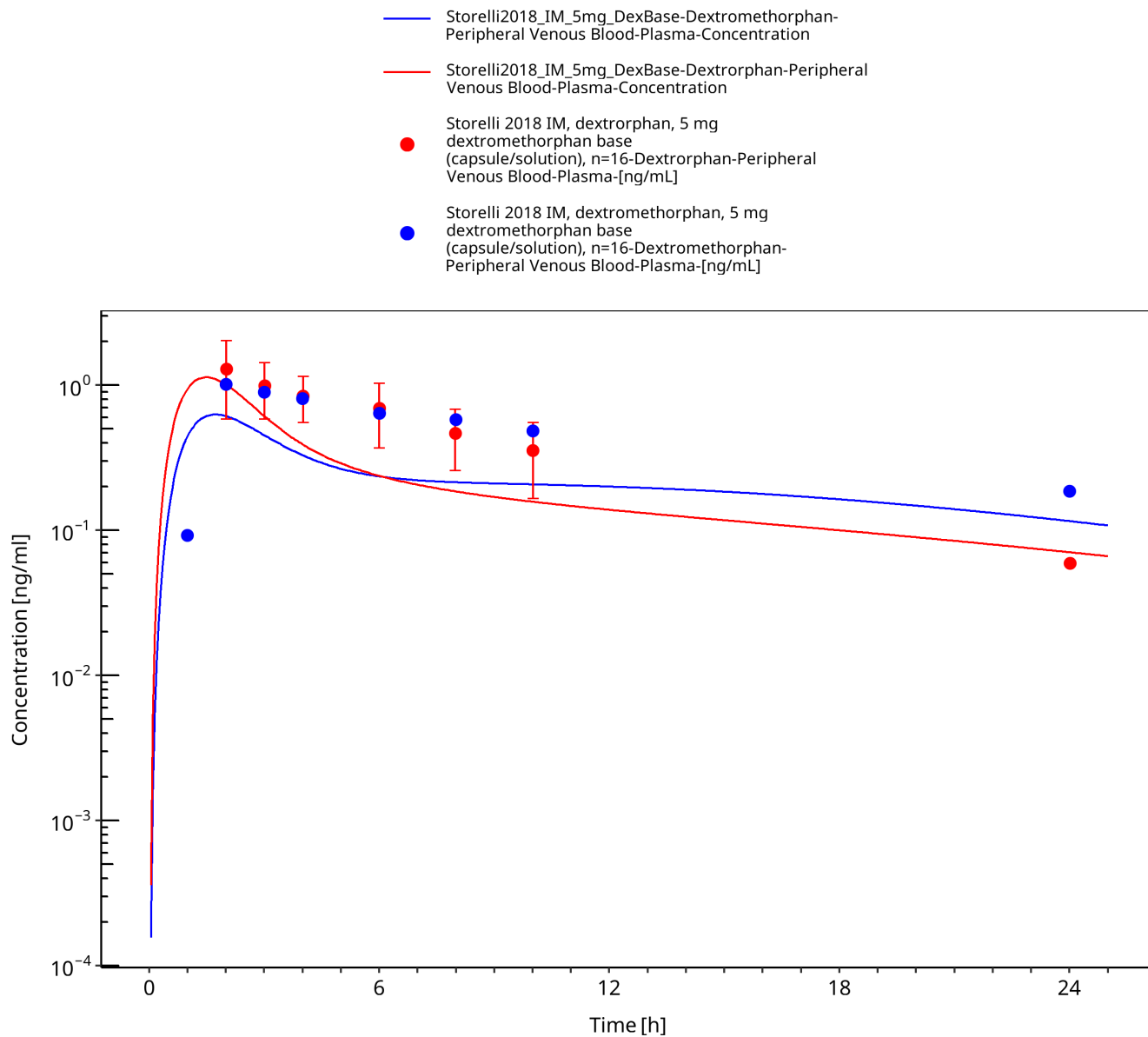


Figure 3-30: Storelli 2018 IM, 5 mg dextromethorphan base (capsule/solution), n=16

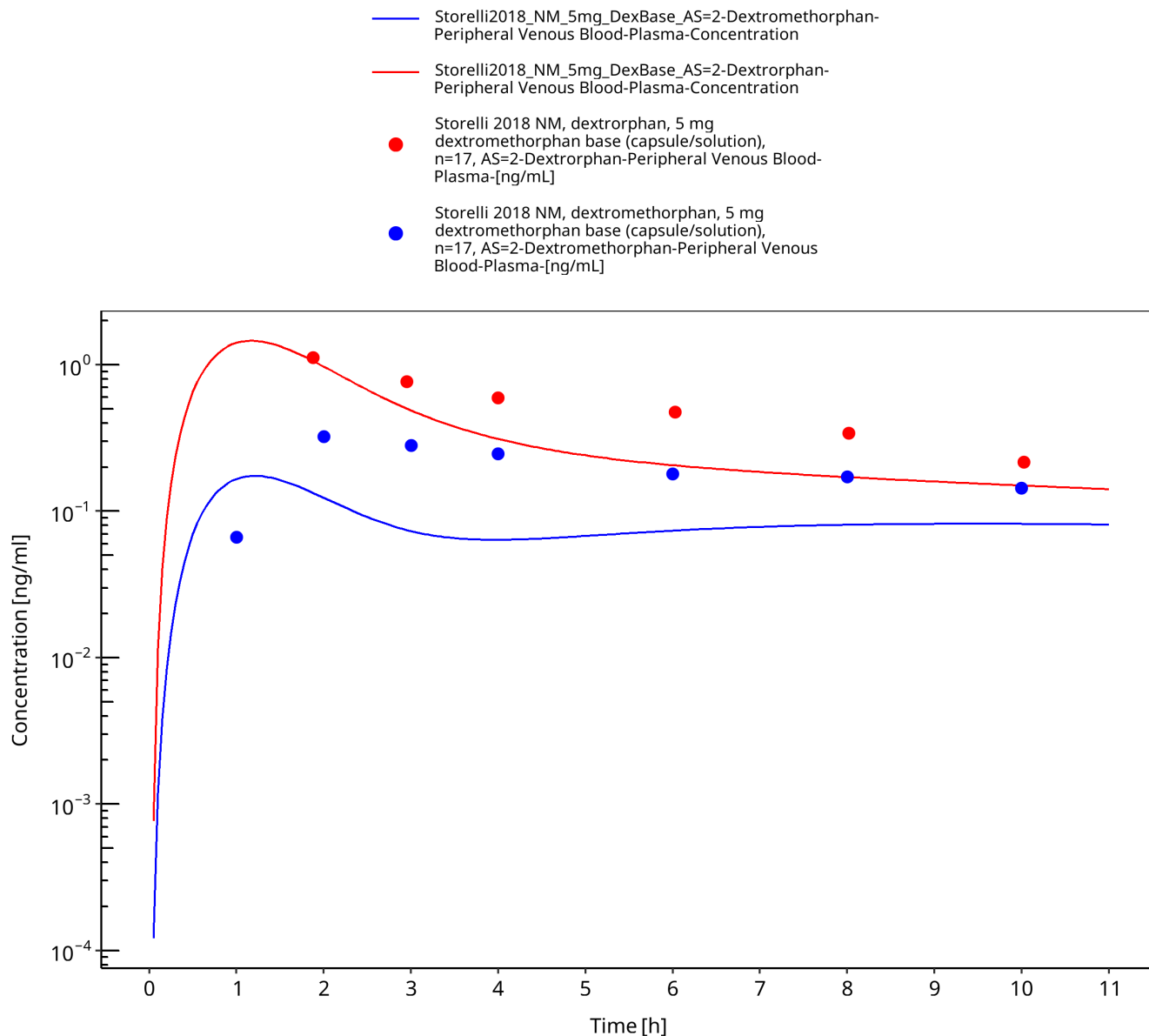


Figure 3-31: Storelli 2018 NM, 5 mg dextromethorphan base (capsule/solution), n=17, AS=2

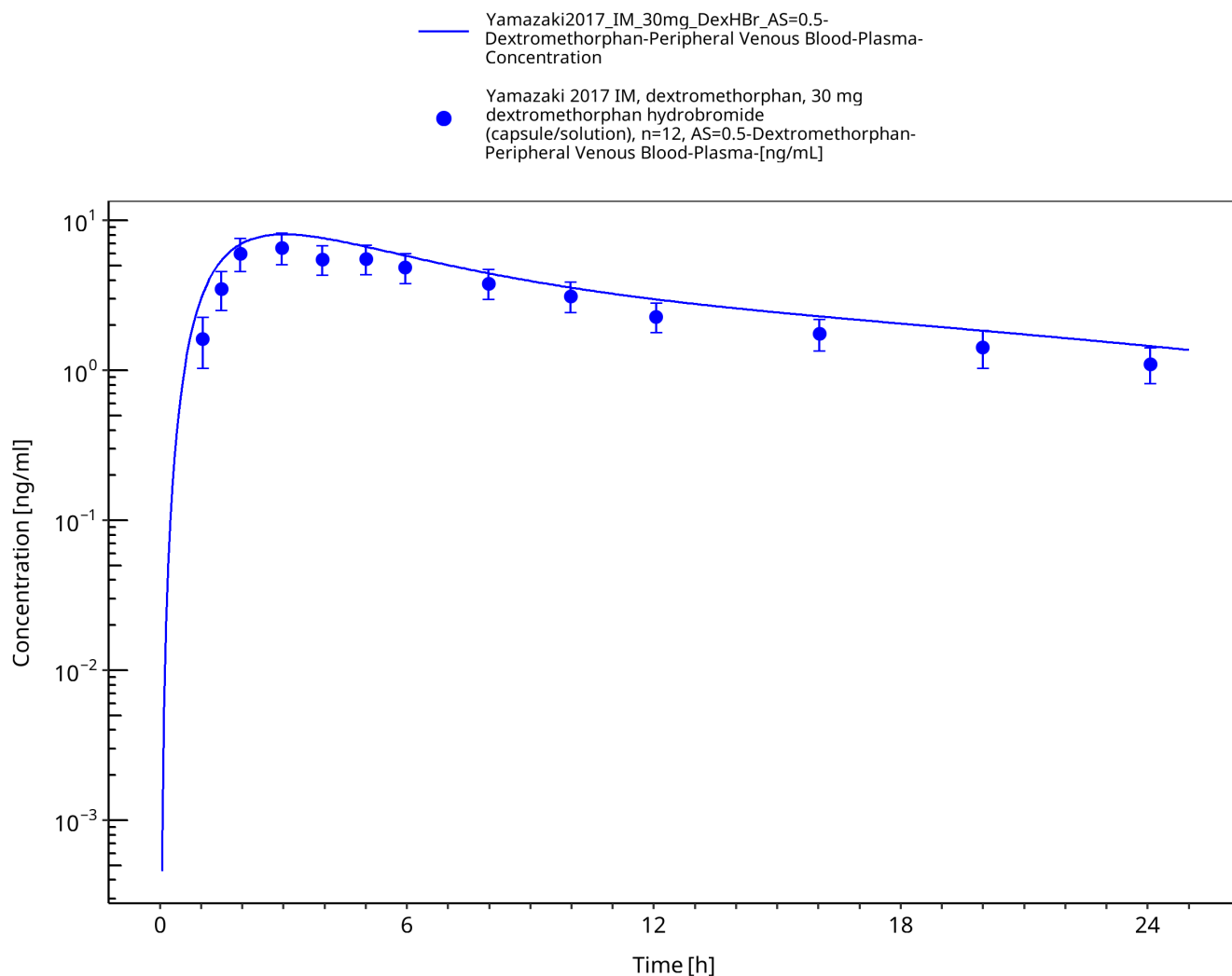


Figure 3-32: Yamazaki 2017 IM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=12, AS=0.5

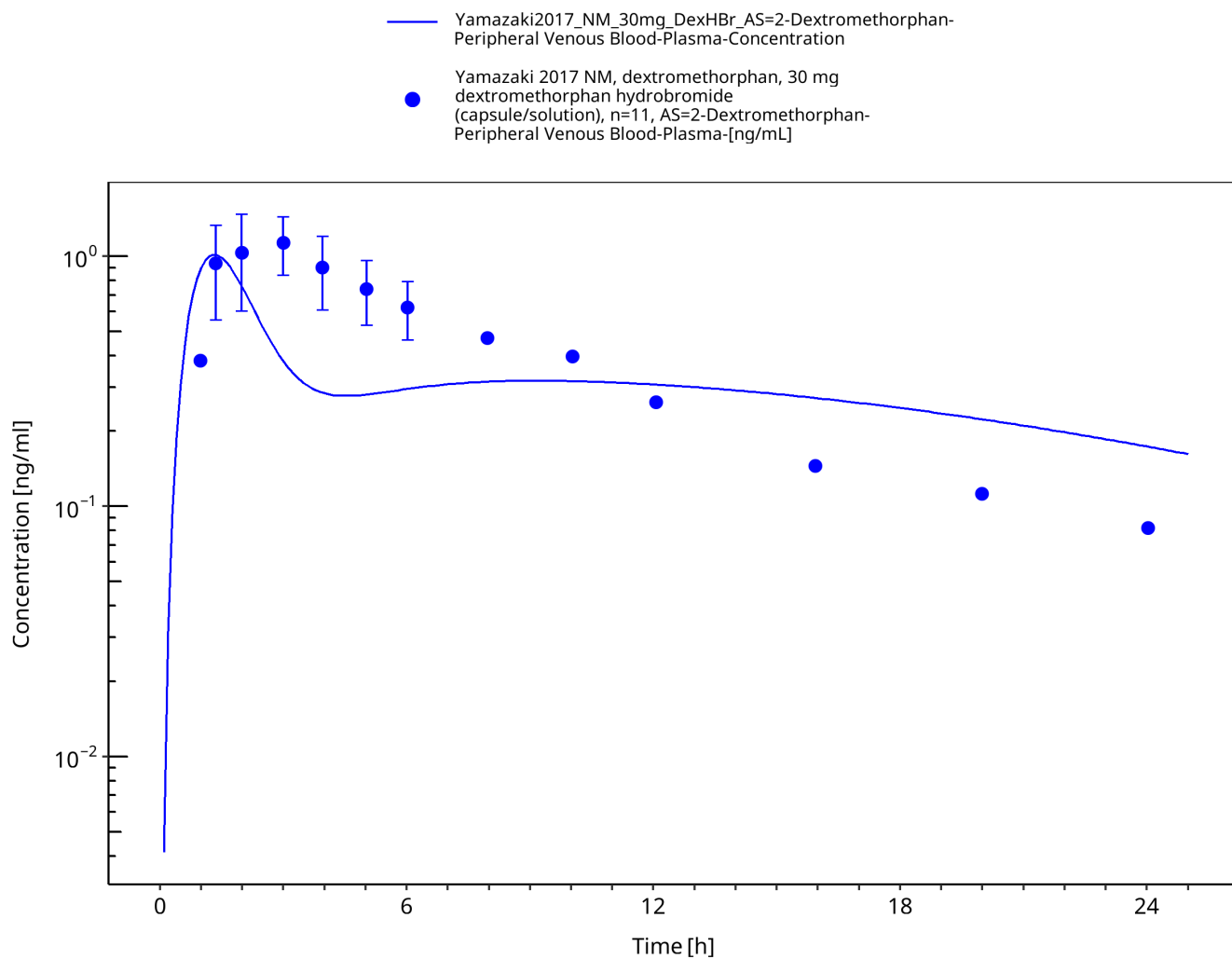


Figure 3-33: Yamazaki 2017 NM, 30 mg dextromethorphan hydrobromide (capsule/solution), n=11, AS=2

## 4 Conclusion

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The herein presented PBPK model adequately describes the pharmacokinetics of Dextromethorphan and its metabolites in adults with different CYP2D6 phenotypes.

In particular, it applies quantitative metabolism by CYP2D6 which can be adapted to a CYP2D6 phenotype using CYP2D6  $k_{\text{cat}}$  activity score scalars. Thus, the model is fit for purpose to be applied for the investigation of DDIs with regards to CYP2D6 metabolism in adults with different CYP2D6 phenotypes.

## 5 References

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**HMDB-a** Human Metabolome Database: Showing metabocard for Dextrorphan (HMDB0060552)

**HMDB-b** Human Metabolome Database: Showing metabocard for Dextrorphan O-glucuronide (HMDB0010341)

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